



CALASIAO COMPREHENSIVE NATIONAL HIGH SCHOOL
Senior High School

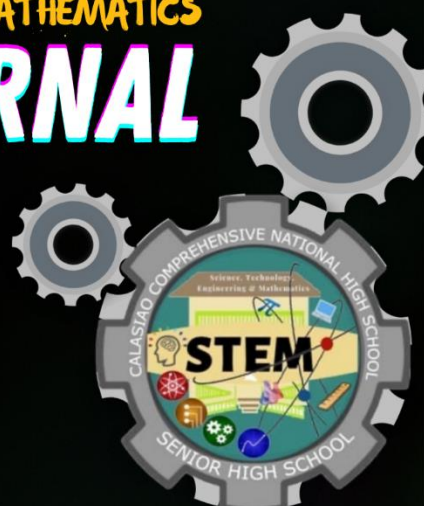
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FOREWORDS

In this continuously growing, improving, and changing world of ours, one thing is for sure: research is the key to true improvement. I was in 8th grade when I first experienced research. An interesting scenario we had back then was that the number of students in my Special Science program class was an odd number, so the decision was either to have one group of three or for one student to work solo on the research. I took on the challenge of working solo because I believed that I would learn better and focus more on the project itself rather than worrying about my research partner.



Since we were new to research back then, we were tasked with doing a science investigatory project proposal. I took a unique approach to this task. Most of my classmates were focusing on biology, while I focused solely on robotics and chemistry. I had a lot of approved research titles, but one outshined the others: I chose a project using stannic oxide as a primary component for making carbon monoxide detectors. We had our proposal defense at the end of the school year, with one of the panel of judges being Engr. Merian Galang, a chemical engineer who would soon be my chemistry teacher. Her expertise was convenient since my research was based on chemistry. Surprisingly, I was awarded best research presenter in our section's defense.

In junior high school, we focused more on the quantitative approach to research. Upon entering 11th grade, we had a new approach to learn: the qualitative approach. It was a very fun experience, connecting words and research. For three years, I had been soloing research, but this time we were required to be in a group. I was tasked with being a leader due to my academic standing in our classroom. With Sir Ivanshane Malala's approach to teaching and implementing, it was a very dynamic and efficient academic experience. I took on the topic of sustainable agriculture as a phenomenology. Since we were only tasked with finishing the proposal, I succeeded in completing the project, although I had to compensate for some things I will not state.

I was looking forward to 12th grade because research was returning to my comfort zone, quantitative research. With Sir Genesis Parel as our research adviser, I was looking forward to a successful school year in research. We were off to a good start with the basics of quantitative research, but on the day of our research groupings, I was late and lost the opportunity to handpick my research groupmates which I see now as the biggest challenge that helped me grow, especially as a team player. In my last years as a senior high school student, so many times did chances to hone our research skills splashed right through our faces. Joining research competitions and teaming up with a solid group (Robert and Dharlyne, and with the very hands-on assistance of Sir Genesis) to work on multiple ideas on a very limited amount of time, we had a great experience (including the opportunity to plant mangroves in RCCC!). It's a truth in life that not everything will go our way. There was a time when we bagged a place (RCCC), a time when we felt obstructed (DNSTF), a time we admitted defeat (DOST SIP), and a time we brought home the gold (NSQ). Indeed, the journey to academic excellence in research is challenging (I even got sick in one of the events while presenting one of our entries in these competitions) but incredibly rewarding. Every obstacle and setback we faced only strengthened our resolve and deepened our understanding. Through perseverance and innovation, we discovered that research is not just about finding answers but about pushing boundaries and creating new possibilities. Reflecting on these experiences, I realized that research is a continuous process of learning, unlearning, and relearning. Each project taught me not only about scientific principles but also about resilience, teamwork, and the importance of adaptability.



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Throughout my experience as a researcher, I recognized that every step, from my initial solo endeavors to collaborative projects, has been a lesson in patience and perseverance. Each failure was a stepping stone, each success a confirmation that the effort was worth it. As I embark on my own endeavors, I will always remember that each challenge is an opportunity for growth---to embrace the unknown, trust in my abilities, and remain committed to my goals. The path of research is not easy, but it is profoundly fulfilling. I hope that when people read this research journal, they will be inspired and realize that our contributions have the potential to drive meaningful change and progress in our world. May it inspire them to pursue research with passion and tenacity, knowing that they can make a significant impact. To the future researchers, let this be a testament to the power of dedication, curiosity, and resilience. See your research not just as an academic requirement, but as a journey of discovery and growth. Whether you work alone or with a team, remember that the true value of research lies in the process of inquiry and the pursuit of knowledge. As we continue to navigate the complexities of the modern world, research remains our beacon of hope and progress. It is through our collective efforts that we can address global challenges and contribute to a brighter future. Embrace the challenges, celebrate the victories, and never lose sight of the impact your work can have. The world needs curious minds and dedicated researchers. Together, let us continue to strive for excellence and innovation, improving our society through the power of knowledge and application.

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Research is a tireless endeavor, requiring the utmost dedication, intellectual rigor, and attention to detail in order to navigate its intricacies, complexities, and challenges. Depending on who you ask, it would mean different things for them. For students, it is one of the challenges that they face in their academic journey. For professionals and researchers, it is a vital tool for advancing knowledge and innovation. For policymakers, it serves as a means of reason for the creation and revision of policies and programs. For me, it became an avenue for learning new things in a new and dynamic field of study, which is Artificial Intelligence, or A.I.

I myself have had previous experience when it comes to research, but while creating RefER as my research paper, I found myself swept away with many new concepts and ideas coupled with expectations of creating a research output that was exemplary and entirely different. I've noticed that past research topics in the STEM strand have revolved mostly around chemistry, bioengineering, and robotics with the use of Arduinos and the like, but not so much around the concept of technology, more specifically when it comes to software programs, and thus, RefER, which is an AI-based chatbot for research article verification, was conceptualized.

In the beginning, the write-ups for the manuscript were more hectic than they needed to be, as we had to make this research from scratch, putting our old research idea behind us and going into uncharted waters, trying to learn concepts such as neural networks, machine learning, and Python as the coding language, which was something that most of my members weren't familiar with.



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Nevertheless, we managed to finish our literature review without any major problems because of their hard work and cooperation. That said, most of the trouble came when we had to create the chatbot itself, which brought out the best as well as the worst in all of us. Many times, we had thought of giving up because of the countless bugs and problems that we had faced, especially in the creation and implementation of datasets. There were also countless hours lost over debugging, only for it to be a missing parenthesis. However, through sheer dedication and understanding of how Python worked alongside other coding languages over time, we gradually made progress as well as improvements to the code and made a functioning chatbot. As simple as it may sound, this is a big accomplishment for us.

Unfortunately, by the time that the chatbot was finished and was ready for testing, little time was left for the training phase, which would mean that it wouldn't be as accurate or as precise as we had perceived it to be. That said, it is to be expected, as chatbots that dabble in neural networks typically take months upon months of training, constant attention, and reconfiguration to ensure the best results when it comes to accuracy and precision. Despite that, we got results that, although they aren't near the standards that we expected for RefER to be reliable, are good enough for having it only be trained for 3 days, which I am satisfied with. There are improvements that can be made to the chatbot; this makes it so that further studies and research may be done in the future, as it is interesting to say the least.

What made this research endeavor memorable aside from the process and measures that were taken into completing it was because of my members. Our group was one with an amiable atmosphere, as we were also friends outside of our research group, which helped immensely in creating a positive and feel-good environment, pushing through challenges together while understanding that not all moments were about making jokes and goofing around and that serious situations require serious input and thinking, which they respected. If it weren't for them, this research would most likely not be possible.

As for Sir Genesis Parel, initially a serious and no-jokes type of person in classes, he turned out to be the opposite outside of classes, being a joyous person, the literal embodiment of "golden retriever energy." Aside from that, he has helped immensely in the creation of our research by correcting and helping to revise our paper to be the best that it could ever be, making the research process less stressful than I thought it'd be and leaving me with the impression that research could also be fun with proper support from the right people. In its totality, the whole research experience was, by definition, a rollercoaster ride, full of ups and downs, twists and turns, and high points and low points. That said, it has given me a lot of valuable insight into group management as well as a sneak peek into the things that I have to learn and work with once I enter college with the course BS Computer Science.

ROBERT ROHAN O. CRUCERO

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Ever since I first stepped into the special science program back in 8th grade, research has been a constant companion on my academic journey. I remember the initial nervousness I felt when our teacher announced we'd be having our very first research project. As the years passed, I found myself gravitating more towards a solo research approach, which came with its own set of challenges.

The transition was not without its difficulties. I recall the countless late nights poring over research, the self-doubt that crept in when my findings didn't align with my expectations, and the



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sheer determination it took to push through each obstacle. But with every setback, I grew stronger, more resilient, and more committed to my research.

When Gio first proposed my name as a leader, I felt a palpable sense of trepidation. The responsibility weighed heavily on my mind, and as I observed the accomplished prowess of my classmates, I found myself wrestling with feelings of intimidation. Imposter syndrome crept in.

The initial quarter of the program proved manageable, as many of my fellow group members displayed a commendable willingness to cooperate. However, as time progressed, I began to encounter significant challenges. Countless nights were spent completing the literature reviews for our manuscripts, often at the expense of attending other classes due to sleep deprivation. It was during these trials that I learned the true value of self-reliance.

The BRAILLEYant project stands out as a particularly unforgettable experience. As we encountered technical difficulties, I found myself harboring doubts about the Orange Pi's RAM capacity, worried that our plans might crumble under the constraints. Yet, we miraculously managed to overcome this obstacle and successfully download the OpenCV library. The moment the camera module functioned as intended, Jake, Jez, and I burst into celebration, high-fiving one another in a display of pure, unbridled joy. This experience solidified my unwavering conviction to approach all projects with a steadfast foundation of prayer. While I was not always entirely content with the level of engagement demonstrated by certain members of my group, I deeply appreciated the genuine respect they displayed, addressing me as "Ate Dhar" despite my not being the eldest among them. This simple yet meaningful gesture held significance for me, a testament to the deep-rooted Filipino values.

Sir Parel, our Science Investigatory Project Adviser, proved to be an exemplary educator whose mentorship left an unforgettable mark on my academic journey. I initially perceived him as reserved and intimidating, but I soon discovered a warm, approachable side to his persona, replete with a keen sense of humor. His fondness for the music of Taylor Swift, especially her song "Welcome to New York," became a memorable aspect of our interactions. This experience has evolved me from a hesitant, self-doubting individual into a confident, resilient leader. I have unearthed an inner strength I never knew existed and have come to recognize the importance of unwavering faith. I am deeply grateful to Sir Gen for believing in me and providing the invaluable guidance and support that enabled me to grow as a leader and an individual. Their mentorship has left a mark on my academic journey and beyond, shaping me into who I am today.

I am filled with a deep sense of gratitude to the Lord, for He has given me this experience that has truly enhanced and enriched who I am as a person.

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Upon entering Senior High School, I was nervous, as we would be having a research subject, and I am someone who did not have a research subject in our lesson. That is why I've never really imagined myself as a research leader as I knew that I could not do so. And I do not have the will and courage as I could barely even handle myself, what more if I lead a 14-person group in a subject I was intimidated by? I remember the nervousness and fear that crept in when it was announced that I would be one of the research leaders in our class. It felt like there was a big stone put on my shoulders and like I was a freshwater fish lost in an ocean.

I was so lost that I didn't even know where to start when making my group's research title. The vastness and complexity of the research world seemed overwhelming, and I was unsure how to navigate this challenging journey. Sacrifices, doubts, and breakdowns became constant companions came along.

The initial stage was full of challenges. Selecting a research topic felt like having to navigate an uncharted course. Discouragement threatened to pull me under when our first three title proposals only lead to dead end. Rejection after rejection left our group frustrated and disappointed with our performance. However, a sheer determination was found when the topic Arbuscular Mycorrhizal Fungi – *Gigaspora margarita* was suggested by our research adviser. I planned to only observe the difference in plant-fungi interaction with the ecto and endo mycorrhizae network. However, other students already come up with it and I was left with observing the factors influencing the its growth – an AMF in arbuscular mycorrhizal in-vitro propagation. I didn't expect that much from our study since I had lost hope that it would become successful and even thought of dissolving our group. Many of our needed equipment and materials were expensive and unavailable in our area, which is why a lot of alterations and improvisations were done to complete the experiment. There were also a lot of errors in our paper one ring of bond paper was not enough so I've always wanted to make sure that all of the parts with needed revisions were near-perfect. Through perseverance and determination, we were able to finish our study and were able to defend it. And we're able to win the Best in Research Paper, making all the efforts worthwhile.

For me, Research is like an ocean. I first deemed it a challenging and complex subject, with only surface-level knowledge. But much like an ocean, it held hidden depths waiting to be explored. The initial surface knowledge felt overwhelming, with challenges that seemed as imposing as rough waves. However, as I explored deeper, I began to appreciate and grow fond of it. While certain challenges persist, the journey holds the potential for exciting discoveries and new insights, experiences, learnings, and friendships that were created and found. The journey, like a deep dive into the ocean, may be daunting at first, but the rewards are more than the worth of the efforts.

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BAHAntay System: Cloud-Based Solar-Powered Early Flood Warning System Using Arduino

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Abstract –Early warning systems are crucial for minimizing flood impacts by providing people with time to evacuate and respond to disasters. The researchers developed a solar-powered, weather-resistant device to enhance flood warning capabilities using Arduino technology and Wi-Fi connectivity. This device will be compatible with cloud-based systems and withstand severe weather conditions. The study proposes an automated method for detecting and predicting floods in Calasiao. It aimed to evaluate the BAHAntay System's performance in water depth measurement and data transmission accuracy compared to manual measurements. The device gained a high accuracy of 99.69% and a precision of 0.73. The system's proven accuracy and reliability make it a valuable tool for enhancing flood monitoring and early warning capabilities. It is ready for use by the community and the researchers recommend its utilization by the local MDRRC unit.

Keywords: solar-powered, Arduino technology, Wi-Fi Connectivity, Water Depth Measurement, Data Transfer

INTRODUCTION

Floods are widely regarded as one of the most detrimental natural disasters in terms of damage to agriculture, private property, and ecology, as they cause significant damage to agricultural lands and disrupt human habitation areas. According to the National Aeronautics and Space Administration (2015), floods are comparable to earthquakes, hurricanes, and tsunamis in terms of various metrics, including but not limited to financial devastation and its impact on human lives. Flooding emerges as a prominent climatic hazard that poses a significant risk to the well-being of individuals and communities. This hazardous phenomenon not only jeopardizes global development prospects but also has the potential to undo the progress made in poverty alleviation and overall development over extended periods (Rentschler et al., 2022). Approximately 2.2 billion individuals, forming 29% of the global populace, inhabit areas that are projected to encounter different levels of inundation during a one-in-a-hundred flood

event (World Bank, 2020). According to the available data from the Emergency Events Database (EM DAT) (2023) and the Centre for Research on the Epidemiology of Disasters (CRED), approximately 60,000 fatalities have been attributed to flood-related incidents from 2013 to 2022.

A report from the Asian Disaster Reduction Center (ADRC) highlights the devastating impact of a shear line cyclone and low-pressure trough in December 2022. It caused significant damage: 52 dead, 18 missing, 16 injured, 681,500 affected people, 4,242 households with partial damage, and 910 houses completely destroyed, with an estimated infrastructure damage cost of 268,565,000 pesos. This underscores the vulnerability of Pangasinan's cities and municipalities to floods. Of the 48 municipalities and towns in the province, 36 are classified as mildly vulnerable or worse, with Dagupan and Calasiao being highly and moderately vulnerable, respectively (Estember & Abiog, 2018).

Climate change and Calasiao's geographical location have heightened its



susceptibility to floods. Low-lying areas such as Mancup, Lasip, and Talibaew, among its 24 barangays, are particularly affected. The recent calamity proclamation by the local government in response to Typhoon Egay and the southwest monsoon highlights the severity of the situation. Initial agricultural damage is estimated at PHP 2.4 million, with an additional PHP 154,000 of high-value crops destroyed. The event impacted 22 houses and caused significant damage to Buenlag Elementary School and Songkoy Elementary School, with estimated damages ranging from 300,000 to 350,000 pesos (Austria, 2023).

Communities facing the persistent threat of flood damage lack adequate resources and strategies to mitigate its consequences effectively. Factors contributing to this vulnerability include insufficient resource allocation, a lack of awareness about flooding, and flawed disaster response procedures.

To address these challenges, communities must develop comprehensive flood risk management strategies, integrating early warning systems. Early warning systems provide timely alerts and enable swift evacuation and response measures. Researchers propose a solar-powered, weather-resistant device with Arduino technology and Wi-Fi connectivity to enhance flood early warning systems. Compatible with cloud-based systems and capable of withstanding severe weather conditions, this device offers an effective and affordable automated solution for flood detection and prediction. It aims to replace manual measurement techniques currently used by local governments, providing a reliable and accurate method for mitigating flood-induced loss of life and property.

METHODS

The BAHAntay System underwent testing in a controlled setting under a post-test-only experimental design, where flood conditions were artificially simulated to evaluate its performance. This method focused on the system's operational

capabilities, particularly its accuracy in measuring water depth and effectiveness in delivering real-time warnings.

To create the early flood warning system, the researchers relied on key components, notably the Arduino UNO R4 Wi-Fi module, which acted as the central control unit. This module, with its Wi-Fi features, managed data processing and sharing. The researchers used connecting wires to ensure smooth communication between system parts.

Additionally, incorporating a solar panel charger was a strategic decision to provide the system with sustainable and renewable energy, thereby improving its autonomy and operational lifespan. Including a waterproof ultrasonic distance measuring module was crucial for accurately measuring water levels, significantly enhancing the system's ability to issue timely and precise flood warnings.

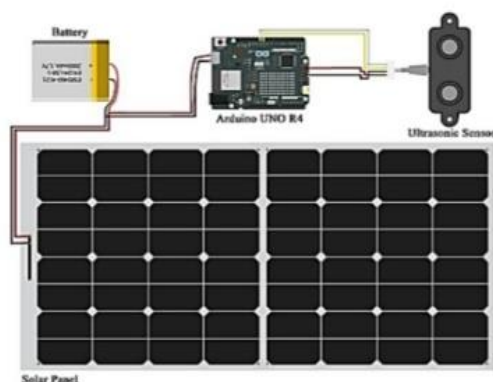
The researchers carefully used these materials to construct and operate the early flood warning system, which used Arduino technology as its foundational framework. The assembly process involved a comprehensive series of steps, including connecting, programming, and integrating the components, resulting in the development of an effective mechanism for early flood detection and alert distribution.

The first step in assembling the system involved gathering the necessary components. These included the Arduino UNO R4 Wi-Fi Microcontroller, the system's main control unit, and the waterproof ultrasonic sensor for automated and accurate water level measurement. A 15W solar panel paired with a 20,000 mAh battery and integrated regulator controller was chosen for a stable power source. Selecting a suitable casing depends on the expected environmental conditions at the device's deployment site.

After assembling the circuitry, connecting to the Arduino IoT Cloud was essential. This involved creating an Arduino Cloud account to link the device to the

broader Arduino IoT ecosystem, transitioning from hardware assembly to software configuration and programming. Connecting the Arduino UNO R4 Wi-Fi to the Arduino Cloud required physically connecting it to a computer and installing the necessary software packages using the Arduino application's user-friendly interface. This setup enabled configuring the Arduino UNO R4 Wi-Fi for internet connectivity via the Arduino Device Setup Configuration. A detailed schematic diagram illustrating the necessary connections for completing the circuitry was meticulously prepared and presented in Figure 1.

Figure 1. BAHAntay System's Schematic Diagram



After uploading the code to the microcontroller, the cloud variable needs to be configured on the Arduino Cloud website as a read-only float variable named distanceCLOUD. Once the software development is complete, the device should be protected within an acrylic enclosure housing the Arduino UNO R4 Wi-Fi, battery, and regulator or controller. A suspension wing should be added to properly position the measuring component with precise leveling to ensure accurate measurement output. The solar panel should be positioned in an elevated area to maximize energy production efficiency.

While the machine's design may vary depending on the location, it typically follows the construction depicted in Figure 2.

Figure 2. BAHAntay System's Current Prototype



The researchers calculated the approximation error to determine the device's accuracy and precision for its standard deviation.

Lastly, the researchers conducted a paired samples t-test to test the null hypothesis, comparing the means of the system's readings to manual water depth measurements.

RESULTS AND DISCUSSION

The table below summarizes the results:

Table 1. Accuracy and Precision of BAHAntay System

Actual Water Depth Measurement (ft)	Average Reading per Trial (ft)	Percent Error (%)	Accuracy (%)	Precision (σ)
1	1.0033	0.57%	99.43%	0.58
2	2.0031	0.36%	99.65%	0.70
3	2.9956	0.33%	99.73%	1.13
4	4.0088	0.24%	99.77%	0.66
5	5.0025	0.10%	99.90%	0.59
	Weighted	0.32	99.69%	0.73

Note: if $\sigma = 0$, "Perfect Performance"; $< = 1.25$, "Acceptable Performance"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≤ 2 , "Unacceptable Performance"

As the table above shows, the device demonstrated exceptional measurement accuracy in all 50 trials (Weighted Mean: 99.69%) with consistently low percent error (Weighted Mean: 0.32%). Precision, measured by standard deviation, remained consistently acceptable (Weighted Mean: 0.73, "Acceptable"). Meanwhile, Table 2 shows the null hypotheses results to confirm the differences between the readings made by the BAHAntay System and the actual water depth measurement:



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Table 2. Paired Samples t-Test Results

t	t-crit	df	p-value	Decision
-1.26	2.01	49	0.21	Not Rejected

Statistical analysis revealed no significant differences between the BAHAntay System's readings and manual water depth measurements (p-value: 0.21 > alpha: 0.05).

CONCLUSION & RECOMMENDATION

The BAHAntay System has demonstrated exceptional accuracy and acceptable precision in measuring water depth, yielding similar, if not identical, results to actual water depth. Therefore, the device is ready for manufacture and implementation. Agencies responsible for disaster management may consider implementing the BAHAntay System in flood-prone areas. Therefore, relevant authorities should initiate the deployment of the BAHAntay System to improve flood preparedness and response efforts.

Regular maintenance and calibration are crucial for keeping the BAHAntay System accurate. Responsible agencies must allocate resources and staff for ongoing maintenance to ensure its effectiveness over time. Further research should aim to improve the BAHAntay System, especially in dealing with Wi-Fi connectivity issues. Collaboration among academic, research, and industry partners can help develop better sensor technologies and data transmission methods, enhancing the system's performance. Future studies may consider adding more sensors or advanced data processing to improve the system. Exploring real-time data analysis and decision support systems could also help improve flood warnings and disaster management.

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RefER: Artificial Intelligence⁵-Based Chatbot for Research Article Verification

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Abstract - Fact-checking and verifying whether or not a research article exists have become an exceedingly arduous and lengthy process due to the rise in scholarly publications in various disciplines in the research sphere, which prompted the researchers to develop RefER, an artificial intelligence-based chatbot for research article verification. The researchers evaluated RefER's accuracy and precision by subjecting a sample ten times to the chatbot for it to distinguish. The sample consisted of a total set of 20 research articles, divided equally between real articles web-scraped from Google Scholar and JSTOR and fake articles from ChatGPT. It used neural networks to enhance data precision by gaining insights from datasets facilitating forecasts. The current prototype has 45.50% accuracy and the test exposed decision-making process inconsistencies in its verdicts, indicating poor precision. These observations underscore the necessity for further refinement and optimization of its algorithms. RefER represents a promising solution for article verification; its current performance highlights the imperative for ongoing development efforts. Enhancements in training protocols and code optimization are essential to bolster its effectiveness in accurately identifying factual research articles.

Keywords: Artificial Intelligence, Research Article Verification, Chatbot, Neural Networks

INTRODUCTION

With the vast number of scholarly publications in various disciplines in the research sphere, fact-checking and verifying whether or not a research article exists has become exceedingly arduous and lengthy.

Furthermore, with the prevalence of applications trained using large language models to create generative artificial intelligence (AI) and Language Models for Dialog Applications (LaMDA) (Collins & Ghahramani, 2021), it has become easier to use these tools to create nonexistent research articles by combining factual and fabricated information based on their data sets (Alkaissi & McFarlane, 2023).

As such, the researchers have

decided to create a solution to help alleviate this problem: RefER, an AI-based chatbot for research article verification made with neural networks built using the Python programming language. Doing so would streamline the process of fact-checking references and, to an extent, citations and information within submitted research papers in academia, as well as for teachers, students, and future researchers looking to find a practical solution to verifying reference points much more efficiently, saving time.

This study aimed to create and test RefER, an AI-based chatbot for research article verification. Specifically, it aimed to determine its accuracy and precision in



verifying research articles. Moreover, it sought to determine whether or not there are statistically significant differences in the conclusions made by the chatbot in research article verification between real and fake articles.

METHODS

The study followed a pre-experimental research design, with a one-shot case study as its experimental model since the researchers can directly assess RefER's performance by comparing it against a hypothesized mean difference, and its application was readily observable without before or after results comparison. A sample dataset consisting of 20 research articles, 10 of which are real and 10 of which are fake, was used to determine the accuracy and precision of RefER in its research article verification. The interpretation of the results was evaluated using a one-sample t-test.

The chatbot was created with Python as its base programming language and the PyTorch library to implement the learning language models (LLMs) to be used in the training of RefER. Functions such as `trl`, which was used for training transformer language models, parameter-efficient-fine-tuning (PEFT) methods for efficient adaptation of the pre-trained models and downloading of datasets, and functions that further create parameters to determine better the correct response, namely Low-Rank Adaptation of Large Learning Models Configuration (LoRAConfig) and Supervised Fine-tuning Trainer (SFTTrainer).

In training the chatbot, an open-source web-scraping code was taken from GitHub. It takes research articles from JSTOR and Google Scholar and lists them in the American Psychological Association (APA) 7th Edition to create the training dataset for the real articles. As for the fake articles were created using prompts from ChatGPT asking for fake research articles to be created in the same format as the real research articles. Three different verdicts were made: (a) "REAL" for cases in which the input given by the user is an actual

research article, (b) "FAKE" for cases in which the research article given by the user is not real, and (c) "CANNOT BE DETERMINED" for cases in which the research article is not a research article, is not in the correct format, which is APA, and for cases in which it is in a language other than English or a non-Latin alphabet.

It was then trained for 72 hours after the proper parameters, such as LoRAConfig, and SFTTrainer, were inputted and after the tokenizer model was implemented, where it did so uninterrupted to ensure familiarity with the possible patterns in both types of research articles.

The frontend was created using Hypertext Markup Language 5 (HTML5) and Cascading Style Sheets 3 (CSS3) for better webpage organization. It included the primary elements for testing the RefER chatbot, such as a textbox in which the articles were to be put and a response section in which the chatbot replied to the given prompt.

The 20 sample research articles were created similarly to the training data for the chatbot. The 10 real articles were randomly taken using web-scraping code from GitHub that takes research articles from JSTOR and Google Scholar. The ten fake articles were created using ChatGPT with the prompt, "Create ten fake research articles in different fields of study from existing journals with nonexistent DOI links written in APA 7th edition reference format, respectively."

Each of the 20 research articles was tested 10 times, and the number of correct verdicts was taken in order to determine the accuracy of the formula:

$$a = \frac{\text{number of correct verdicts}}{\text{number of trials}} \times 100.$$
 After a collective 200 trials, the accuracy of each sample was listed down, and the mean accuracy was determined.

Meanwhile, the researchers calculated the standard deviation of its accuracy per sample to describe its precision.

The researchers evaluated if the chatbot could attain the target by verifying



accuracy and precision by testing the null hypotheses using a one-sample t-test. The null hypotheses should not be rejected to achieve the desired outcome: having the accuracy and precision of the mean values equal to 100%. Otherwise, if the null hypotheses are rejected, the chatbot's accuracy and precision should be within 90% to be somewhat reliable as a reference verifier.

RESULTS AND DISCUSSION

The researchers developed RefER to improve research verification and help future researchers, teachers, and students. The study aimed to evaluate RefER precision and accuracy through testing. The results of this testing, based on 20 data points, are summarized in Table 3 of the paper.

Table 3. *Verifying Accuracy and Precision of RefER*

Parameter	Mean	t	p-Value	Decision
Verifying Accuracy (%)	45.50	-1.73	<0.01	Rejected
Verifying Precision (σ)	36.92	-1.73	<0.01	Rejected

Note. If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"

The test revealed that the current prototype performed poorly, with a low verifying accuracy of 45.50 % and an unacceptable verifying precision of 36.92. The null hypothesis was not rejected by the one-sample t-test, which showed a significant difference in p-value < 0.01 from the expected mean values for these parameters. These findings imply that RefER's functionality needs redesigning and reprogramming, which the researchers attribute to insufficient training time for the chatbot. The study shows development opportunities, especially when considering network architecture, hardware resources, optimization, and dataset size.

CONCLUSION & RECOMMENDATION

Further improvements on the chatbot are needed, especially concerning its code, training optimization, and time spent in training. Given the preceding conclusions, the researchers recommend that future researchers look to further the capabilities and performance of the RefER chatbot may increase the training time used in the study, preferably within the 1–6-month range, so that it may be more accurate and precise in its verdicts regarding research articles in general. Furthermore, the researchers encourage future study researchers to collect more trial data points and increase the number of trials, increasing the chatbot's accuracy and precision in the data collection phase.

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BRAILLEyant: Orange Pi-based Braille Translator with Optical Character Recognition and Voice Output

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Abstract – Traditional Braille displays have been widely used to aid people, primarily those with visual impairment and disabilities, but they can be expensive and bulky, limiting their accessibility and portability for many individuals with visual impairments. To address this gap, the researchers pioneered the BRAILLEyant system, an Orange Pi-based system that utilizes an affordable and compact single-board computer platform to provide a more accessible Braille translation and output solution. This study aimed to assess the performance of the BRAILLEyant system in terms of optical character recognition (OCR) accuracy and precision, as well as the accuracy and precision of the voice output and Braille displays feature tested under a one-shot case study. The system achieved a high accuracy rate of 97.6% for voice output and 99.2% for Optical Character Recognition. It was also precise in performing the parameters, with a mean of 0.09 and 0.37, respectively. The null hypotheses were not rejected, which means that there is no significant difference between the predetermined text specified mean and BRAILLEyant in terms of accuracy, indicating that the standards were met by the device on the test measured by the researchers.

Keywords: Braille, Orange Pi, Optical Character Recognition, Voice Output, Assistive Technology

INTRODUCTION

Vision, as the most dominant of our senses, plays a critical role in various aspects of life. However, visually impaired individuals face challenges in acquiring knowledge, mobility, literacy, and employment opportunities. The prevalence of visual impairments has been steadily increasing globally, with an estimated 2.2 billion people affected worldwide (World Health Organization, 2023). In the Philippines, nearly 4.62 million individuals suffer from visual impairment, including 1.18 million with cataract-related issues (Montemayor, 2022).

Braille has been instrumental in enabling blind and visually impaired individuals to independently read books,

documents, and other written materials, promoting inclusivity and active participation in society (Zatserkovnyi et al., 2019). However, learning Braille can be challenging, and Braille products' high cost and bulkiness limit their accessibility and portability.

The Orange Pi-System presents a promising avenue for cutting-edge research, particularly in addressing the research gap identified by the researchers. Their pioneering work explores the potential of Orange Pi-based devices as Braille displays tailored for individuals with visual impairments. This endeavor aims to bridge the gap between technology and accessibility, revolutionizing how visually-impaired individuals access and interact with



information.

To address the challenges mentioned, the researchers developed an affordable and locally adaptable technology called BRAILLEYant—a Braille translator with optical character recognition (OCR) and voice output based on the Orange Pi platform. This study aims to evaluate the effectiveness of the BRAILLEYant system in providing accessible Braille translation through OCR and voice output.

This study aims to assess the effectiveness of BRAILLEYant—an Orange Pi-based Braille translator with optical character recognition (OCR) and voice output. Specifically, the study aims to evaluate the performance of the OCR feature in terms of accuracy and precision, as well as the accuracy and precision of the voice output and Braille display. Additionally, the study seeks to determine whether there are statistically significant differences between the accuracy of BRAILLEYant's OCR and voice output/Braille display features and a perfect accuracy of one hundred percent. By addressing these objectives, the research aims to contribute to advancing assistive technologies for individuals with visual impairments.

METHODS

The study employed an experimental research design to investigate the effectiveness of an Orange Pi-based Braille translator using Optical Character Recognition (OCR) in enhancing precision and accuracy for visually impaired individuals. A one-shot case study was chosen as the experimental model, considering the presence of only one group in the dataset used to test the null hypotheses. The study utilized predetermined text samples, consisting of five characters each, to evaluate the effectiveness of the BRAILLEYant system. A one-sample t-test was conducted to test the null hypotheses.

The core hardware component was the Orange Pi PC H3 board, the

9 microcontroller. The system incorporated the Orange Pi Camera 5mp OV5640, a compatible camera module, to capture real-time images of predetermined text. Direct Current 5V push-pull solenoids connected to L298N motor drivers were utilized for the Braille displays. On the software side, Ubuntu was installed as the operating system for development, along with VScode and Python libraries. This involved integrating OCR software or libraries and developing and implementing algorithms for translating recognized text into Braille.

The installation involved downloading and flashing the Ubuntu operating system onto a 32GB microSD card. Python was installed as the primary programming language, and Visual Studio Code (VSCode) was used as the integrated development environment (IDE) for coding the BRAILLEYant system. The hardware integration included the Orange Pi as the central computing platform, a high-resolution camera module for capturing text images, a refreshable Braille display for tactile output, and a speaker or audio jack for text-to-speech functionality.

The coding process of BRAILLEYant began with initializing video capture using the OpenCV library. The Tesseract-OCR engine was then employed to convert the text within the captured frame into a string. The system determined the corresponding Braille character for each recognized character based on a predefined mapping. Simultaneously, it activated the corresponding Braille display pins on the hardware interface. This process of capturing video frames, recognizing text, translating to Braille, and providing audio and visual feedback continued in a loop until the user or an external event terminated the program.

To test the BRAILLEYant system, the researchers utilized predetermined text samples, each consisting of five characters. The system translated these samples five times, and the frequency of correct OCR recognition, accurate display, and correct voice output was recorded separately. The



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collected data was then used to calculate accuracy and precision for optical character recognition and voice output. The experiment included a total of five text samples.

The researchers utilized a quantitative approach to evaluate the accuracy of the BRAILLEYant system in assisting individuals with visual impairments in reading predetermined text. Accuracy was measured by comparing the number of correctly read texts to the total number of texts read, expressed as a percentage. The formula used for accuracy calculation was:

$$\text{Accuracy (\%)} = (\text{Correct Verdicts} / \text{Total Texts Read}) \times 100$$

A value of 1 was assigned if the system correctly interpreted the text, while 0 was assigned if the system could not determine the correct interpretation. The researchers conducted five trials per text sample, recorded the verdict (correct or incorrect) for each trial, and tallied the total correct and incorrect verdicts for each sample. The accuracy percentage was then calculated for each text sample. To assess reading precision, the researchers calculated the standard deviation.

RESULTS AND DISCUSSION

Table 1 summarizes the results of testing BRAILLEYant's Optical Character Recognition (OCR) for accuracy and precision. The system achieved perfect accuracy in all five trials, with a mean accuracy of 99.2% and a precision of 0.09, indicating high accuracy and low variability

Table 1. Optical Character Recognition Accuracy and Precision of BRAILLEYant

Test Characteristics	Mean Accuracy (%)	Precision (σ)
1	100	0.00
2	100	0.00
3	100	0.00
4	100	0.00

5	96	0.45
Weighted Mean	99.2%	0.09

Note: Each Test Characteristics consisted of 5 letters tested for five trials each.

If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"

10 Table 2 presents the accuracy and
10 precision testing results for BRAILLEYant's Voice Output and Braille Displays. The system achieved a consistent accuracy of 96% in the first three trials and perfect accuracy in the fourth and fifth trials. The mean accuracy across all trials was 97.6%, with a standard deviation of 0.27, indicating relatively low variability and high precision.

Table 2. Voice Output and Braille Displays Accuracy and Precision of BRAILLEYant

Test Characteristics	Mean Accuracy (%)	Precision (σ)
1	96	0.45
2	96	0.45
3	96	0.45
4	100	0.00
5	100	0.00
Weighted Mean	97.6%	0.27

Note: Each Trial has 5 Characters Tested

If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"

The one-sample t-tests conducted for OCR, Voice Output, and Braille Displays did not reject the null hypothesis. The p-values were 0.37 and 0.07, respectively, suggesting no significant difference between the observed and expected performance.

Table 3. One-Sample t-Test Results

Parameter	t	t-crit	p-Value	Decision
Optical Character Recognition	-1	2.78	0.37	Not Rejected



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Verifying Precision (σ)	-2.45	2.78	0.07	Not Rejected
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The findings indicate that BRAILLEYant's Optical Character Recognition (OCR) achieved high accuracy, with all characters correctly recognized in each trial. The mean accuracy was 99.2%, demonstrating consistent and reliable performance. The Voice Output and Braille Displays feature also exhibited high accuracy, with a mean accuracy of 97.6% across all trials. While there was some variability in the accuracy results, the system performed consistently. The one-sample t-tests confirmed that the observed performance of both OCR and Voice Output and Braille Displays was close to the expected performance, supporting the system's reliability. Overall, these results highlight the effectiveness of BRAILLEYant in accurately recognizing Braille characters and providing output, making it a valuable tool for individuals with visual impairments.

CONCLUSION & RECOMMENDATION

Having passed the standards set on the parameters that the researchers tested BRAILLEYant with, the braille translator exhibited excellent performance, with accurate and precise Optical Character recognition, voice output, and Braille displays. While the BRAILLEYant braille translator demonstrated excellent performance regarding accurate Optical Character Recognition (OCR), voice output, and Braille displays, there is still room for further improvements. The researchers acknowledged that the device's functionality could be enhanced to provide an even better user experience. Further studies may focus on enhancing the machine's field of view, word pronunciation, and line segmentation.

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TALKING PLANT: ARDUINO-BASED SOLAR-POWERED WATERING SYSTEM WITH SMS NOTIFICATION

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Abstract - In the era of smart technology, people seek automation to simplify tasks. Manual watering, while common, can be inefficient and labor-intensive, leading to water waste and reduced crop productivity. As a solution, the researchers embarked on developing a solar-powered watering system equipped with SMS notification capabilities that utilize Arduino technology with the purpose of enhancing gardening for better plant care and monitoring. The device was designed to provide plants with precise and accurate watering, with the additional feature of notification through SMS for accurate and precise monitoring. Through systematic experimentation and comprehensive data analysis, the research evaluates the reliability of SMS notifications and assesses the impact of the Talking Plant treatment on various plant characteristics. Notably, the study achieves a 100% accuracy and precision rate for SMS notifications, affirming their effectiveness in facilitating timely communication between the device and users. Plants treated with the Talking Plant show significant improvements in key growth parameters compared to those using manual watering methods, including increased height, better leaf quality, thicker stems, and more leaves. These findings underscore the system's potential to revolutionize plant care, providing an efficient solution for enhancing plant health.

Keywords: Plant care, Solar-powered, Arduino, SMS Notification, Watering system

INTRODUCTION

Many find fulfillment in nurturing plants, but maintaining their health can be challenging due to busy lifestyles. Automation is sought to simplify tasks, with manual watering proving inefficient and labor-intensive. Automated systems ensure precise watering, which is crucial for plant health. Insufficient watering hinders photosynthesis, while excess water can harm plants (Kathpalia & Bhatla, 2018). Real-time soil moisture monitoring is vital for sustainable agriculture. The Talking Plant system addresses these challenges by utilizing Arduino and SMS notifications. It combines soil moisture sensors and SMS alerts for precise plant care, enhancing

automation and human interaction in gardening. The Talking Plant system integrates Arduino technology with SMS notifications to provide accurate plant care. It aims to develop a low-cost, dependable solution for efficiently monitoring soil moisture levels and watering plants. Combining automation with human interaction facilitates better plant care and treatment, enhancing the gardening experience. This innovative approach utilizes modern technology to ensure optimal plant health and productivity while catering to the needs of plant enthusiasts.

The study compares the characteristics of plants subjected to Treatment 1, utilizing the Talking Plant



system—a solar-powered watering system with SMS notification, and Treatment 2, which involves traditional or manual watering methods. It assesses the height, stem thickness, and number of leaves of plants under each treatment. Additionally, the accuracy and precision of the Talking Plant system in sending SMS notifications are evaluated. Statistical analysis is conducted to determine if there is a significant difference in the characteristics of plants between the two treatments, indicating the effectiveness of the device in monitoring plant health.

METHODS

The study employed a true experimental design with a two-group, posttest-only approach to evaluate the efficacy of the Talking Plant, an Arduino-based solar-powered watering system with SMS notification. *Capsicum annum* seedlings were carefully selected based on optimal health indicators such as vigorous growth and uniform appearance. The experimental group received treatment with the Talking Plant system, while the control group underwent traditional watering methods. Data collection involved measuring plant height, stem thickness, and the number of leaves to evaluate growth metrics (Voxco, 2021).

For data collection, the researchers utilized measuring tape, vernier calipers, and leaf area meters to obtain accurate measurements of plant characteristics. The subjects, consisting of *Capsicum annum* seedlings, were carefully monitored under controlled experimental conditions to ensure uniformity across treatment groups (Solomon et al., 2019). Data analysis was conducted using independent sample t-tests to compare mean differences in plant height, stem thickness, and the number of leaves between the experimental and control groups.

Additionally, the study incorporated an SMS notification procedure, where the Talking Plant system sent alerts to users based on soil moisture levels. The system was

programmed to emit sound notifications, send SMS alerts, and activate the water pump based on predetermined conditions (Suntoyo et al., 2015). The accuracy and precision rates of the SMS notification system were evaluated to assess its effectiveness in real-time monitoring and communication. Through systematic data collection and statistical analysis, the study aimed to provide insights into the potential of the Talking Plant in enhancing plant growth and health metrics, while also assessing its reliability in automated plant care management.

RESULTS AND DISCUSSION

The table below summarizes plant measurements across treatment groups A (Talking Plant) and B (Control Group), including height, stem thickness, and leaf count. These metrics offer insights into the effectiveness of the Talking Plant compared to traditional methods, highlighting its potential to improve plant health and productivity.

Table 1. Measurement of Plant Characteristics

Parameter	Treatment Groups							
	A (with Talking Plant)				B (Control Group)			
	1	2	3	Mean	1	2	3	Mean
Height (cm)	27	26	27	26.70	19	22	23	21.30
Stem Girth (mm)	2	2	2	2	1.5	1.6	1.8	1.63
Number of Leaves	38	36	32	35.3	27	19	29	25

As seen in Table 1, The group monitored with the Talking Plant System grew better in all parameters measured (height: $26.70 > 21.30$; Stem Girth: $2.00 > 1.63$; and Number of Leaves: $35.3 > 25.00$), compared to the control group; suggesting that the System did its job efficiently.

Meanwhile, the table below summarizes the independent samples t-test results on the parameters regarding the plant characteristics.

Table 2. Independent Sample T-test Result for the Experimental Group



Parameter	p-Value	Decision
Height (cm)	0.013	Rejected
Stem Thickness (mm)	0.014	Rejected
Number of Leaves	0.043	Rejected

As seen in Table 2, The null hypothesis is rejected because the p-value is less than the chosen significance level, indicating that there is sufficient evidence to support the alternative hypothesis, meaning that there is a significant difference or effect present, confirming the findings.

CONCLUSION & RECOMMENDATION

Talking Plant is an Arduino-based solar-powered watering system with an SMS notification designed by the researchers. The main goal of Talking Plant is to use an Arduino as a microcontroller board to combine soil moisture sensors with SMS notifications for better plant monitoring and plant treatment. Several experiments and observations conducted by the researchers proved the device's reliability for keeping the plants healthy and hydrated as the plants under this treatment group grew taller, more robust, and bearing more fruit. The accuracy and precision rate of the Talking Plant produced 100% in testing the SMS notification, guaranteeing reliable performance in monitoring and watering plants with precise SMS notifications. Based on the given results, there is still a need for improvement on the device Talking Plant for extensive watering experience.

Researchers should improve the device's programming to reduce errors during testing or experimenting. According to the findings of the Talking Plant study, this action is important to increase the device's functioning and usability. Future devices like the Talking Plant should be refined to be thinner and more polished as soon as possible. This will improve plant comfort and reduce interference, as indicated in the

study's findings on the device's usefulness even when dealing with minor watering issues. Transitioning from SMS-triggered to perimeter-triggered alerts, as exemplified in the Talking Plant, can provide more proactive monitoring in real-time. This upgrade, while optional, has the potential to improve the device's monitoring capabilities and user experience dramatically. Further studies on the long-term effects of the Talking Plant or similar devices on plant growth and health may provide insights into the long-term usefulness and benefits of such devices in plant care.

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NOTIFUR: ARDUINO-BASED ANIMAL TRACKER WITH DISTRESS MONITORING FEATURE THROUGH GLOBAL SYSTEM FOR MOBILE COMMUNICATIONS

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Abstract - Globally, technologies have been invented to scope potential advantages and provide effective tools for any aspect of life, including animals. Monitoring efforts inefficiency is challenging as various difficulties may interfere with observations due to animals that tend to hide, are unapproachable, and have wild tendencies. In this sense, inventing innovative systems such as tracking devices and distress monitoring will enable humans to detect abnormalities in the health of an animal in a way that is accessible and safer. The researchers developed an Arduino-based animal tracker with advanced distress features through a global system for mobile communication that contributes potentially to revolutionizing animal care practices and seeks to contribute to the field of wildlife conservation with components connected to the Arduino nano board and integrated with programmable code for triggering SMS. The assessment of tracking and distress accuracy has resulted in a notable accuracy of its measurements in all test subjects across trials, having NotiFur shown acceptable accuracy and precision in tracking but performs poorly in distress signal accuracy, which leads to the researchers to recommend better components to ensure better results that will take inspiration from this study.

Keywords: Arduino, animal tracker, monitoring, distress, SMS, wildlife

INTRODUCTION

In wildlife monitoring, conservationists face various difficulties in observing animals. Some tend to hide, cannot be approached by a human due to their wild tendency, and are affected by humans, which impacts their behavior and mental state.

Using tracking and monitoring for animals can benefit animals and

conservationists in monitoring even at a distance. Monitoring efforts by wildlife conservationists have enabled a better understanding of the occurrence, distribution, and status of wildlife in conservation.

According to Morrant et al. (2022), tracking affords researchers the opportunity to analyze animal movement data using sophisticated statistical methods. In some



instances, it also allows researchers to visualize and follow the same paths as their study animals, viewing the landscape similarly.

The researchers conducted a study to explore the functionality of Notifur while assisting other researchers in exploring a similar scope. The researchers hence designed NotiFur, an Arduino-based animal tracker that detects the location of the animal where the device is attached and monitors the pet's heart rate with distress monitoring through a global system for mobile communication (GSM).

METHODS

The components used were as follows: (1) Arduino-nano microcontroller; (2) NEO-6M GPS tracker, a GPS receiver chipset responsible for receiving signals from GPS satellites and processing them to collect and calculate the device's position; (3) MAX30102 heart rate monitor, a photodetector transmitted through the tissue or the skin, measuring animal's heart rate (beats per minute) based on the intensity of infrared light transmitted through animal tissue; (4) SIM800L GSM module, the last component to make up the device NotiFur, operating at quad-band which supports compatibility for different cellular networks worldwide, and it also contains a SIM-card slot which will take advantage of the collection of data. The NotiFur is powered by an Arduino-nano connected to a 5v battery. The microcontroller was designed for flexibility using an open-source Arduino Integrated Development Environment (IDE) software. The code was adjusted to change the notification mode, allowing for customization based on user preferences or any specific application required.

After creating the device, the researchers tested its performance under

a one-shot case study, a pre-experimental research design. Attaching the device to three test subjects (dogs), they measured its tracking accuracy and ability to identify distress signals by comparing its readings for ten trials on the actual location and status parameter according to the heart rate monitor. They also measured its battery life.

The researchers determined the total duration of the battery life by calculating using the formula $\text{Battery Life} = B - A$, where point B was the end time, observed from the digital clock, and point A was the starting time, where the NotiFur has been turned on or used. They calculated the accuracy percentage in tracking location and distress using the formula $A = \frac{\text{number of correct readings}}{\text{total number of trials}} \times 100$, where the total number of trials conducted is 10, within a test subject consisting of three (3) animals, all with the same qualification in a single group. They then calculated the mean accuracy on both parameters. Meanwhile, the precision of the device's readings will be assessed by calculating the standard deviation of the measurements. Finally, the researchers used a one-sample t-test to test the study's null hypotheses.

RESULTS AND DISCUSSION

The table below shows the results from testing the average accuracy of NotiFur's location and distress monitoring acquired from three test subjects.

Table 1. Results from Testing Average Accuracy

Test Subjects	Tracking Accuracy	Distress Signal Accuracy
1	80%	80%



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2	100%	90%
3	90%	70%
Mean Accuracy	90.00%	80.00%
Precision (σ)	10.00	10.00
Test Subjects	Tracking Accuracy	Distress Signal Accuracy

Note: Each Test Subjects were observed for ten trials to determine accuracy; Standard Deviation Interpretation: 0.0= perfect comparison; 1.25= acceptable; 1.25-1.49= acceptable to marginal performance; 1.5-1.99= marginal performance; 2.0= unacceptable performance.

Comparing the mean values of the tracking and distress signal accuracy shows a concerning difference. NotiFur is accurate in terms of tracking accuracy, collecting a computed value of 90% across three (3) test subjects; however, in terms of distress accuracy, NotiFur had only collected a computed value of 80% across three test subjects, which suggests that there is a need for improvement of the device. The precision of the NotiFur is concerning; the researchers computed a precision value of 10%, and in terms of the SDI (Standard Deviation Index) value of interpretation, this value suggests that the NotiFur shows an unacceptable performance. The researchers attributed it to the fact that the experimental setup lacked test subjects, which they considered a flaw in their procedure. According to Bruggermann and Bizer (2016), experiments with small sample sizes suffer from issues related to generalizability and external validity. Such studies often produce results that may not apply to a broader population because they are more susceptible to sampling variability and other biases. Regardless, the researchers take the result as an indication that their prototype has yet to achieve the desired outcome for the measured parameters. For the results of the battery life, the researchers have

determined a total of three (3) hours of uptime until the battery is fully drained, which they collected by activating the device the whole day and using the formula Battery Life=B-A, in this case, Point B is denoted as nine (9) am, marked by the researchers as the NotiFur's starting time, Point A which is denoted as twelve (12) pm, where after 3 consecutive hours of activation, the NotiFur's battery life is depleted. Throughout this duration, the NotiFur remained continuously active and operational, actively collecting data until asked for the data. The timeframe provides an overview of NotiFur's performance under consistent usage, allowing researchers to measure the device's power consumption over time. The computed value of three hours is a bad indication of the NotiFur's performance, as it is only limited to 3 hours, a very short time for a device dedicated to tracking and monitoring an animal's location and state of being.

Finally, the table below summarizes the results of the hypothesis testing under a one-sample t-test via JAMOV:

Parameter	Mean	t	p-Value	Decision
Tracking Accuracy	-1.73	4.30	.22	Not Rejected
Distress Signal Accuracy	-3.46	4.30	.07	Not Rejected

Despite having different results per test subject, the researchers still failed to reject the null hypothesis, as the p-value is significantly higher than the significance level of the null hypothesis 0.05. The reason behind this could be the flaw in the experimental procedure they have previously cited, which is the lack of test subjects. A small sample size can lead to several issues: it reduces the study's



statistical power, making it harder to detect a true effect if one exists. This increases the likelihood of a Type II error, where the null hypothesis is not rejected even though it is false (Bacchetti, 2010).

CONCLUSION & RECOMMENDATION

NotiFur, in the parameters of tracking accuracy, has shown accurate findings but lacks in the distress monitor, which suggests that NotiFur is still subject to improvement. as for its precision parameters, there is a need for re-evaluation as there is a flaw in the experimental setup by the researchers. NotiFur needs to be tested on multiple test subjects, which is true considering that the researchers lacked test subjects, severely affecting the results. The battery life of the NotiFur is also subject to improvement, as its total battery life is three hours at best. Hence, NotiFur is still subject to prototyping and further studies, as very few studies are relative to the device.

The researchers also recommend having a longer time frame for conducting and prototyping the device, seeking professional guidance for informative insights, transitioning from SMS-triggered to parameter-triggered to provide flexibility, which improves accuracy and efficiency, to invest in better components to lessen the rate of running into malfunctions, and lastly, to implement a power button for future replications so that the battery will not be running continuously and lessen the rate of battery drainage.

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SOLARSCARE PRO: ARDUINO-BASED SCARECROW WITH SOLAR TRACKING AND AUTOMATIC FERTILIZER APPLICATION FUNCTION

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Abstract - To improve crop protection and productivity, the researchers developed the SolarScare Pro system, an Arduino-based scarecrow with solar tracking and automatic fertilizer application capabilities. They employed a one-shot case study, a form of pre-experimental design, to evaluate the system's effectiveness in detecting and discouraging pests, mainly birds and animals, to prevent crop damage. Data gathering entails assessing sun-tracking accuracy, motion-detecting capabilities, and autonomous fertilizer application efficacy, analyzed quantitatively. The results show high sun tracking and motion detection accuracy, with significant gains in autonomous fertilizer delivery ($t\text{-Stat} = 2.44$, $t\text{-Crit} = 1.83$, $df = 9$, $p\text{-Value} = 0.036$). However, there are certain drawbacks, such as the inability to schedule fertilizer applications and the limited range of motion detection. Future studies should focus on improving Arduino technology integration, including scheduling capabilities for fertilizer applications, improving motion detection algorithms, and developing methods for scarecrow arm movement and alarm triggering.

Keywords: *Arduino Uno, solar tracking, automatic fertilizer, pest control, motion detection*

INTRODUCTION

Crop damage poses a significant challenge that keeps people from meeting their food demands (Sonal et al., 2021). Strategies to enhance crop production and mitigate losses are essential for achieving maximum productivity and income, particularly in countries like the Philippines, where agriculture plays a crucial role in livelihoods (Cagasan and Dogello, 2021).

The effectiveness of the traditional scarecrow has decreased to the extent that some farmers believe it to be useless (Król-Agriculture, 2020). Moreover, the depletion of fossil fuels and their detrimental impact on pollution and climate underscores the importance of transitioning to renewable alternatives like solar energy. Embracing solar and other renewable sources offers a cleaner and more environmentally friendly

approach to meeting our energy needs (Tiwari et al., 2016). Additionally, fertilizer is any organic or inorganic material, natural or synthetic, that contains one or more of the chemical components essential for plant growth. By applying fertilizer, with modest increases as needed, leaching can be prevented, and nitrogen absorption can be boosted (Utah State University, 2019).

Therefore, the researchers developed SolarScare Pro, an Arduino-Based Scarecrow with Solar Tracking and Automatic Fertilizer Application Function that deterred birds and other animals that could harm crops. This eco-friendly invention ensured proper plant growth while providing a dependable and environmentally friendly option. The study has evaluated the effectiveness of this innovative system in enhancing crop protection and productivity, rendering it a



dependable and environmentally conscious choice for farmers.

METHODS

The SolarScare Pro system comprises an Arduino Uno R3 microcontroller board, solar panel, gears, linkages, two arms, and poles to construct the scarecrow's movement mechanism. The LDR sensors and servo motor enable sun tracking, and PIR sensors detect pests, triggering scarecrow deterrent devices. The system's automatic fertilizer application ensures timely nutrient supplementation, set to start at a specified time. After programming the Arduino Uno using IDE software, the device is secured in an enclosure containing the circuits. The system is tested and monitored for proper functioning and long-term crop protection.

The research employed a one-shot case study, a pre-experimental design, to evaluate the effectiveness of the device, measuring its performance on the following parameters: Solar Tracking, Motion Detection (by Distance), and Automatic Fertilizer Application. Solar tracking allows the device to adjust its position in response to changing sunlight levels throughout the day. The researchers observed the solar panel's position to see if it met the standard position at a specific time. If the position was right, the researchers put 1, otherwise 0, in the observation sheet. This ensures the device can effectively harness solar energy to power its operations and maintain its functionality.

The researchers also observed if it works at a specific distance; they put 1, otherwise 0, in the observation sheet to see if the motion detection still works at a specific distance. Comparative analysis assessed accuracy by comparing sensor results to manual measurements of three distinct treatments. At the same time, statistical analysis evaluates precision using measures of central tendency and variability. Validation under different conditions and visualization of data trends aid interpretation, leading to conclusions about the sensor's

performance and potential recommendations for improvement.

Similarly, the researchers tested the automatic fertilizer in different trials; they put one if it works, otherwise 0, in the observation sheet. The data analysis findings are summarized in a comprehensive report, interpreting results in the context of system performance and practical implications for use.

The researchers tallied the results per trial then calculated the accuracy per observation in the parameters using the formula:

$$\text{Accuracy} = \frac{\text{total number of correct reading}}{\text{total number of observations}} \times 100$$

They then measured the precision by calculating the standard deviation per parameter.

Finally, they subjected the observations from the parameters measured to a One-Sample t-test, comparing the data to the hypothesized mean accuracy of 100%. If the researchers failed to reject the null hypotheses, the device passed the standard in performing the functions. Otherwise, rejection of the null hypotheses may indicate the need to further enhance the robot in performing functions. It may however be deemed sufficient if it at least gained 95% accuracy in performing the parameters tested.

RESULTS AND DISCUSSION

The table below provides a summary of the results obtained from the testing process:

Table 1. Solar Tracking analysis

Time	Trials										Accuracy (%)
	1	2	3	4	5	6	7	8	9	10	
6:00 am	1	1	1	1	1	1	1	1	1	1	100
9:00 am	0	1	1	1	1	1	1	1	1	1	90
12:00 pm	0	1	1	1	1	1	1	1	1	1	90
3:00 pm	1	1	1	1	1	1	1	1	1	1	100
Mean											95%
Precision (σ)											5.77

If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"



Regarding its Solar Tracking Accuracy, it garnered a mean accuracy of 95% but with poor precision (5.77 = Unacceptable Performance). However, it is also important to note that its accuracy slightly decreased between 9:00 am and 12:00 pm. The slight decrease in accuracy between 9:00 am and 12:00 pm may be due to changing sunlight conditions or mechanical wear of the tracking system. According to National Geographic Education (2023), factors like cloud cover or environmental conditions could also play a role. Cloud cover affects weather forecasting by influencing sky conditions, precipitation predictions, and regional temperatures. Regardless, the results indicate favorable performance for the SolarScare Pro in Solar Tracking.

The table below summarizes the results:

Table 2. Motion Detection by Distance

Distance (meters)	Trials										Accuracy (%)
	1	2	3	4	5	6	7	8	9	10	
1	1	1	1	1	1	1	1	1	1	1	100
2	1	1	1	1	1	1	1	1	1	1	100
3	1	1	1	1	1	1	1	1	1	1	100
4	0	1	1	1	1	1	1	1	1	1	90
Mean											94%
Precision (σ)											8.94

If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"

SolarScare Pro also garnered a favorable performance in Motion Detection, with a mean accuracy of 94% but with poor precision (8.94 = Acceptable Performance). While it could detect motion in all distances measured, its accuracy decreased by 4 meters, which could be attributed to factors such as changes in lighting conditions, environmental obstructions, or limitations in sensor sensitivity. Ansari (2024) emphasizes that regular monitoring and maintenance ensure sensors function optimally and remain unaffected by environmental factors. This involves conducting periodic checks specific to each sensor type and implementing routine cleaning procedures to remove debris.

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Further investigation is needed to pinpoint the exact cause of the decreased accuracy at this specific distance. Overall, the device performed better in its function.

Meanwhile, the table below summarizes the performance of the device on the Automatic Fertilizer Application:

Table 3. Automatic Fertilizer Application

Trials										Accuracy (%)
1	2	3	4	5	6	7	8	9	10	
1	1	1	1	1	1	1	1	1	1	60%
Precision (σ)										0.52

If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"

As seen in Table 3, while still under acceptable performance (precision = 0.52), it indicated poor precision in the Automatic Fertilizer Application, garnering only 60% Accuracy. Rubio and Maz (2020) highlight that the necessity for improvement arises from the difference between the precision score, indicating overall acceptable performance, and the low accuracy rate, specifically in the Automatic Fertilizer Application. This indicates that although the robot performs adequately in certain aspects, it faces challenges in accurately performing this function. Therefore, enhancing the robot's ability to apply fertilizer automatically would improve its overall performance and effectiveness.

Lastly, the table below presents the One-sample T-test results of Solar Tracking, Motion Detection by Distance, and Automatic Fertilizer Application.

Table 4. One-Sample t-Test Results

Parameter	t	t-crit	p-Value	Decision
Solar Tracking Accuracy	-1.73	3.18	0.18	Not Rejected
Motion Detection Accuracy	-1.5	2.78	0.21	Not Rejected
Automatic Fertilizer Application Accuracy	2.44	1.83	0.0	Rejected



As seen in Table 4, except for the Fertilizer Application, the researchers failed to reject the null hypotheses in the parameters, garnering p-values greater than an alpha level of 0.05 (Solar Tracking Accuracy: 0.18; and Motion Detection Accuracy: 0.21). This confirms that while SolarScare Pro performed accurately in solar tracking and motion detecting, it needs further enhancement in its coding for performing automatic fertilizer application.

The SolarScare Pro has shown to be highly accurate in Solar Tracking (95%) and Motion Detection (94%). However, it has not demonstrated the same level of accuracy as the Automatic Fertilizer Application, with a score of 60%. The SolarScare Pro system performed all measured functions precisely, with Solar Tracking at 0.22, Motion Detection at 0.24, and Automatic Fertilizer Application at 0.52. The only significant difference in performance was observed in the Automatic Fertilizer Application ($0.04 > 0.05$), indicating a need for further improvements in this area. Overall, the SolarScare Pro has demonstrated impressive capabilities and potential for enhancing agricultural practices.

CONCLUSION & RECOMMENDATION

The researchers successfully designed SolarScarePro, an Arduino-Based Scarecrow with Solar Tracking and Automatic Fertilizer Application Function that deterred birds and other animals that could harm crops and an eco-friendly invention that ensured proper plant growth while providing a dependable and environmentally friendly option. Their tests revealed that it was accurate and precise in Solar Tracking and Motion Detection of up to 4 meters, but enhancements can still be made to increase its performance on the said functions further. However, its programming still needs major enhancement, particularly in the automatic fertilizer application. Nevertheless, the robot is functional and can

be readily used to serve the purposes of its creation.

22 The researchers recommend improving Arduino integration, upgrading motion sensors, and optimizing scarecrow movement for better pest control. Researchers suggest enhancing the scheduling to the fertilizer function and motion detection algorithms. Future researchers must focus on dynamic arm movement and incorporating alarms for increased effectiveness in crop protection.

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ARDUINO-BASED HYDROPONIC FARMING WITH VEGETABLE FERMENTATION IN WATER AS NUTRIENT SOURCE

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Abstract - Agriculture plays a significant role in the Philippine economy. The study investigated the comparative efficacy of Arduino-based hydroponic farming versus traditional methods, which is crucial in addressing the pressing need for sustainable agricultural practices. The research uses sensors, weighing scales, calipers, and rulers to evaluate both systems' plant characteristics, humidity, water temperature, and pH levels. While statistical analysis reveals no significant disparity in plant characteristics (p -value=0.79; $t = 2.45$), Arduino-based hydroponics demonstrates superior environmental control, particularly in humidity ($p = <0.01$; $t = 22.19 > 2.57$) and water temperature regulation ($p = 0.024$; $t = 3.53 > 2.78$). These findings underscore the complexity of transitioning to innovative agricultural technologies amidst sustainability challenges, highlighting the importance of holistic assessments for informed decision-making of farming practices. However, conclusive evidence for its superiority over traditional hydroponics still necessitates further examination, especially concerning cost-effectiveness and maintenance.

Keywords: hydroponics, humidity, water temperature, plant characteristics

INTRODUCTION

Agriculture accounts for around 40% of the labor force in the Philippines and adds, on average, 20% to the Gross Domestic Product. About 70% of all agricultural output is accounted for by agricultural businesses, which is the primary source of this output. Philippine agriculture is distinguished, among other things, by its poor productivity and lack of diversity (Brown et al., 2020). Unlike traditional farming, the hydroponics system can produce vast amounts of food with relatively little space and does not require soil to grow food. This method involves growing plants on artificial or natural substrates so the roots can easily draw nutrients from a prepared nutrient solution (Velasquez-Gonzales et al., 2022). The fundamental idea behind a hydroponic NFT system is a technique for growing plants with

shallow roots and circulation, which allows the plants to receive enough water, nutrients, and oxygen. The roots of the plants are immersed in water that contains nutrient solutions and are continuously pumped by a pump as they grow in layers of polyethylene. When a plant's rooting region develops and grows in a nutrient solution to the point that the top of the root is visible between the nutrient solution and styrofoam, oxygen may be fully absorbed and is sufficient for normal growth (Iswanto et al., 2020).

Plants in a hydroponics system, such as lettuce, cabbage, and basil, grow very rapidly due to the natural dissolved nutrients processed by vegetable fermentation in water as a nutrient source. A very new method of plant cultivation called soilless cultivation is utilized almost entirely in greenhouses and



involves a nearly fully controlled environment (Fussy & Papenbrock, 2022).

METHODS

The coconut fiber, sourced from reputable suppliers, underwent hydration after being compressed into blocks, followed by thorough rinsing to eliminate excess salts. Optional sterilization was conducted by heating the blocks to deter pests and diseases. Subsequently, the coco peat was prepared and moistened before filling the containers for plant transfer. For traditional hydroponic farming, a square configuration of styrofoam with cup-holding holes was arranged, and nutrient-rich solutions were prepared and administered to seedlings in cups filled with coco peat. An aeration system was incorporated using an air stone, and environmental considerations included low sunlight exposure. A PVC structure, integrated with a water pump connected to a reservoir, facilitated even water distribution through PVC pipes. Net cups filled with coco peat were placed in the PVC holes, and the Arduino-controlled watering system was programmed and sensor-equipped for monitoring and adjustments.

The study utilized descriptive and inferential statistical methods to explore distinctions between Arduino-based and traditional hydroponic systems across the outlined variables. The researchers observed the plant characteristics in their hydroponic systems using the scale developed by the Bureau of Agriculture and Fisheries Products and Standards (BAFPS) and the Philippine National Standard for Lettuce (PNS) in 2005.

RESULTS AND DISCUSSION

The table below summarizes the comparison of the plants under the two hydroponic systems prepared by the researchers:

Table 1. Tabulated Mean Value of the Plant Characteristics

Hydroponic System	Leaf Quality	Plant Length	Stem Girth	Dry Weight	Crop Yield	Total Mean
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Traditional	4.29	5	2.76	3.71	3.24	3.8
Arduino-Based	4.43	5	3	3.91	3.52	3.97

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The two systems were discovered to satisfy the specifications specified in the Philippine National Standard for Lettuce (PNS) in 2005 across all aspects. Nevertheless, Arduino-based hydroponic farming demonstrated a slight superiority, indicating a minor advantage over the alternative system.

Table 2. Hypotheses Testing Results on the Plant Characteristics of the Hydroponic Systems

Parameter	t	t-crit	p-value	Decision
Leaf Quality	-0.70	2.02	0.48	Not Rejected
Plant Girth	1	2.09	0.32	Not Rejected
Stem Girth	-1.74	2.09	0.09	Not Rejected
Dry Weight	-0.8	2.02	0.43	Not Rejected
Crop Yield	-0.69	2.02	0.49	Not Rejected
Total	0.27	2.45	0.79	Not Rejected

All of the null hypotheses were not rejected for the parameters, cementing the previous interpretation. Meanwhile, the table below summarizes the humidity and water temperature comparison on the hydroponics:

Table 3. Humidity and Water Temperature Data in the Hydroponic Systems

System	Mean Humidity per Day (%)			Weighted Mean	Mean Water Temperature per Day (°C)			Weighted Mean
	1	2	3		1	2	3	
Traditional	62.7%	65.3%	67.3%	68.21%	27.2°C	27.7°C	28.1°C	28.1°C
	68.5%	69.3%	71.2%		28.3°C	28.7°C	28.8°C	
Arduino-Based	72%	77.2%	78.4%	77.85%	26.1°C	26.3°C	26.3°C	27.5°C
	79.6%	79.8%	80.1%		27.4°C	27.4°C	28.8°C	

The Arduino-based system exhibited a humidity differential of 9.64%, which may have affected the plant. Additionally, there was a slight decrease in water temperature by 0.6°C compared to the weighted mean, it could impact the growth.

Table 4. Hypotheses Testing Results on the Humidity and Water Temperature of the Systems

Parameter	t	t-crit	p-value	Decision
Humidity	22.19	2.57	<0.01	Rejected
Water Temperature	3.53	2.78	0.02	Rejected



Both of the parameters indicated rejection of the null hypotheses, emphasizing the difference on the humidity and water temperature between the two hydroponics.

CONCLUSION & RECOMMENDATION

Despite achieving comparable lettuce production results to traditional hydroponics, the Arduino-based hydroponic farming version demonstrates effectiveness attributable to plant characteristics. Nonetheless, the Arduino-based system shows promise for further development and improvement, particularly in optimizing humidity and water temperature. Enhancements to the prototype, such as augmenting water pump capacity for better circulation, optimizing plant spacing, and fine-tuning nutrient solution levels, could elevate the performance of the Arduino-based system.

In future research endeavors, it is imperative to ensure a consistent supply of fermented vegetables immersed in water to meet demand, potentially necessitating replacement every 3-5 days based on the acidity level of the circulating nutrient solution in the hydroponic system. Additionally, increasing the number of water pumps and designing a prototype for water circulation within the reservoir can significantly boost efficiency by enhancing humidity and water temperature regulation.

Innovators should prioritize selecting lettuce varieties renowned for resilience, particularly in hot climates. These varieties, such as green rapid lettuce, can thrive under such conditions when coupled with efficient water circulation.

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SoP'er LIiT Bot: Safety Surveillance for Disaster Risk Reduction

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Abstract - A natural disaster is a highly damaging impact on society or a community following a natural hazard event. Natural disasters are perilous because they cause extensive destruction and affect many individuals. Schools are among the areas profoundly affected by earthquakes, as they usually consist primarily of students who are particularly vulnerable during such events. This led the researchers to develop So'Per LIiTBot, a line-tracking minibot that aims to monitor individuals inside a school or room following an earthquake, significantly aiding in saving lives. The prototype currently exhibited satisfactory accuracy (90%) and acceptable precision (0.94, "Acceptable Performance") in line tracking but slightly slower velocity (0.17 m/s) than usual minibots. The current prototype need further enhancements and reprogramming.

Keywords: Line following robot, natural disaster, schools, earthquakes, monitor

INTRODUCTION

Line Following is one of the most important aspects of robotics. A Line Following Robot is an autonomous robot that can follow either a black line or a white line drawn on a surface of a contrasting color. It is designed to move automatically and follow the line. The robot uses arrays of optical sensors to identify the line, which helps the robot stay on track. The array of four sensors makes its movement precise and flexible (Rahidka, 2023). Typically, a line follower robot consists of two motor sets, which will refer to the left and right motors. Based on the signals from the left and right sensors, respectively, both motors rotate. Moving forward, turning left, turning right, and stopping are the four sets of motions that the robot must execute. When both the sensors are on a white surface, and the line is between the two sensors, the robot should move forward, and both the motors should rotate such that the robot moves in the forward direction. The left sensor is on top of the dark line, whereas the right sensor is on the white

part; hence, the left sensor detects the black line and gives a signal to the microcontroller. Since the signal comes from the left sensor, the robot should turn in the left direction. Therefore, the left motor rotates backward, and the right motor rotates in the forward direction. Thus, the robot turns towards the left side. To turn the robot in the right direction, the left motor rotates forward, and the right motor rotates backward, and as a result, the robot turns towards the right direction. Since both sensors are above the line and can detect the black line simultaneously, the microcontroller is programmed to regard this as a reason to terminate the procedure. As a result, the robot stops moving when both motors are stopped (Rajesh, 2021).

Earthquake is one of our major problems in modern society, it can strike suddenly and without warning. Moreover, The Philippines is home to many natural wonders of the world. It is an archipelago. The Philippines is situated along the basin of the Pacific Ring of Fire. It is between the



Eurasian and Pacific Plate where there is a large number of earthquakes and volcanic eruptions (Maminta, 2019). Earthquakes can disrupt essential services such as water, electricity, and communication networks, making it difficult for emergency responders to reach affected areas and for survivors to receive aid. In some cases, communities may not be adequately prepared for earthquakes in terms of building codes, emergency response plans, and public education about earthquake safety measures. Earthquakes can cause extensive damage to infrastructure, including buildings, roads, and bridges, leading to a threat of lives for individuals, particularly to our emergency responders.

SoP'er LITBot is a Solar-Powered line-tracking minibot created by researchers for Safety Surveillance and Disaster Risk Reduction. It aims to help vulnerable individuals in earthquakes; it can track people under collapsed buildings and perform surveillance on damaged infrastructures due to earthquakes. The robot is equipped with mini solar panels to generate electricity and power its operation, making it eco-friendly and cost-effective. This SoP'er LITBot has a camera that helps in spying or helps the army and other emergency responders to collect information; it can inspect every edge and even enter small hole areas that are not accessible to humans.

METHODS

The researchers tested the minibot under a One-shot case study because the experimental setup does not require pretest-posttest data, and comparison with a control group will not be necessary for measuring the robot's line tracking accuracy, line tracking precision, and velocity. They can easily measure the variables by simply comparing SoP'er LIT Bot performance to a hypothesized mean value.

The researchers gathered the required materials for building a SoP'er LIT Bot. This includes acquiring an Arduino with the size of UNO and a 52 x 52mm solar panel.

They used an infrared ray sensor and other wires that will connect to solar, battery, Arduino UNO, and also a broken power bank with a battery capacity of 2200 mAh to store energy from solar. Lastly, the size of the toy car was 130 x 155mm, and the researchers also used a camera to detect the surroundings. Some materials, like the Arduino UNO, toy car, and solar panel, were provided by instructors assisting the researchers in their study. However, the researchers purchased additional materials such as a camera, power bank, DC motors, and wires.

The researchers created the code in Arduino Uno via IDE. After gathering all the materials needed to build a SoP'er LIT Bot, the researchers attached the Arduino UNO to the sensor and mounted it to a predesign toy car, along with the other wires and the power bank. The solar panel was placed on top of the toy car to serve as a cover. To make a permanent setup, everything should be connected to a jumper wire and secured with electrical tape. The camera was mounted in front to serve as the robot's head. The researchers built a storage box once assembled the SoP'er LIT Bot.

The researchers will have the SoP'er LIT Bot follow the prepared line track for ten trials, and record the number of times it followed the track correctly. If the line tracking is accurate, it is interpreted as "1" and "0" if not. The researchers will also measure the velocity by measuring the time it took to finish the track per trial.

The researchers calculated the line-tracking accuracy using the formula:

$$\text{Accuracy} = \frac{\text{Observed Value}}{\text{Total Number of Observation}} \times 100$$

The hypothesized mean value is set to 100%. Meanwhile, the precision was assessed by calculating the standard deviation of the measurements. The results were interpreted based on the Standard Deviation Index (SDI).

The researchers also computed the velocity of the robot per trial using the following equation:

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$



where the total displacement equals 18.2 meters

The hypothesized mean value is set to 0.43m/second, taking as a basis the threshold set by Wilbrandt (2016). Finally, the researchers tested the null hypothesis via One sample t-test to compare the set standard of SoP'er LiIT Bot to its actual performance. Failure to reject the null hypothesis will indicate a favorable result because it means that the robot met the standard, and its rejection will say otherwise. The researchers used Microsoft Excel to analyze the result quantitatively.

RESULTS AND DISCUSSION

As seen in Table 1, the results obtained by the researcher from the gathered data.

Table 1. Performance of SoP'er LiITBot

Parameter	Score	t	t-crit	P	Decision
Line-tracking Accuracy (%)	90	-3.35	2.26	0.01	Rejected
Line-tracking Precision (σ)	0.94	---	----	----	-----
Velocity (m/s)	0.17	-10.42	2.26	<0.05	Rejected

If $\sigma = 0$, "Perfect comparison with consensus group"; ≤ 1.25 , "Acceptable"; 1.25-1.49, "Acceptable to Marginal Performance"; 1.5-1.99, "Marginal Performance"; and ≥ 2.0 , "Unacceptable"

The accuracy results indicate an average accuracy of 90% across all collected data, meeting almost the hypothesized mean for accuracy. Precision, determined through standard deviation, was found to be 0.94, within the acceptable range of 1.25. This indicates that the minibot also achieved an acceptable precision level. However, the velocity of 0.17 m/s suggests slow movement, falling short of the standard speed for a line tracking robot, which Wilbrandt (2016) sets at 0.43 m/s by default. A one-Sampled t-test also confirmed such findings, suggesting a need for improvement in the programming of the robot to perform the functions evaluated.

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CONCLUSION & RECOMMENDATION

The analysis of the So'Per LiIT Bot's performance reveals satisfactory accuracy and precision, meeting the expected standards. However, its velocity fell short of the typical speed for line-tracking robots, indicating a notable deficiency in movement capabilities. While the robot demonstrates competence in accuracy and precision, enhancing its speed is crucial for ensuring efficient real-world operations.

The effectiveness of the robot in disaster scenarios can be significantly enhanced by incorporating features for obstacle avoidance, stair climbing, and terrain adaptation. While the robot meets standards in accuracy and precision, there is a clear need for enhancing its velocity, which could be achieved through weight reduction to improve acceleration. They suggest deploying the robot for disaster and risk reduction efforts, particularly in inaccessible or hazardous areas, recommending its use to Disaster Risk Reduction and Management Councils. Additionally, they advise future researchers to focus on enhancing sensor accuracy, optimizing weight for improved performance, and exploring designs to facilitate human access to hazardous zones. These recommendations aim to enhance the effectiveness and versatility of the So'Per LiIT Bot in real-world scenarios and future research endeavors.

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ENVIRONMENTAL FACTORS INFLUENCING FUNGAL GROWTH OF *Gigaspora margarita* IN ARBUSCULAR MYCORRHIZAL IN-VITRO PROPAGATION

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Abstract – Fungi have a wide range of contributions to agriculture. They are significant for both sustainable food production and environmental conservation. *Gigaspora margarita*, while a common arbuscular mycorrhizal fungus, is yet to be studied, and due to its potential application for sustainable measures in agriculture and possible fungal evolution when grown in arbuscular mycorrhizal in-vitro propagation, the researchers investigated how different environmental factors influence its growth in the root organ culture method. The study focused on temperature, moisture level, and light intensity in a controlled setting. Results showed that moderate levels of all three factors enhanced the root colonization and hyphal growth of *G. margarita*. Interestingly, higher levels of light intensity produced the highest mean out of all groups. Based on the data analysis and interpretation, all three environmental factors – Temperature, moisture level, and light intensity-influence the growth rate of *G. margarita*.

Keywords: *Gigaspora margarita*, arbuscular mycorrhizal fungi, in-vitro propagation, environmental factors, agriculture

INTRODUCTION

The Philippines is known to be an agricultural country as products obtained in this sector are used in many industries thus, it plays a vital role in the country's economy. However, in recent years, the Philippines faced numerous crises in the agricultural sector such as soil pollution, decreased Gross Domestic Product, and decreased funding for the agricultural sector. Fungi are eukaryotic organisms found anywhere, digesting and absorbing their food through their cell wall (Carris et al., 2012). Fungi thrive in seemingly hostile environments and are the major decomposers of nature. It is also involved in various activities, such as symbiotic relationship, especially mutualism and parasitism in living things and fungi. An example of this is the mycorrhiza network. Mycorrhizal fungi are a subset of fungi that establish mutualistic associations with plant roots. O'Callaghan (2013) cites that

approximately 90% of plants have mycorrhizal partners. Mycorrhizal fungi have various species – the common types are ectomycorrhiza and endomycorrhiza. Ectomycorrhiza is when the fungus produces a sheath around the root, producing hyphae that grow into the root and out into the soil. In contrast, endomycorrhiza does not produce sheath; the hyphae will grow within the cells and in the soil. *Gigaspora margarita* is a type of endomycorrhiza. It is an obligate symbiont that requires being associated with a host plant to accomplish its life cycle (Bonfante, 2022). Its association with plants creates an exchange crucial for the mutual benefit of the fungus and the host plant as it can form an association with dicotyledonous and monocotyledonous plants. Many examples from other microbial systems show that long-term in-vitro propagation may lead to domesticated strains of fungi that may differ genetically and functionally (Kokkoris



& Hart, 2019). *Gigaspora margarita*, being a mycorrhizal fungus, has the potential to have a fungal evolution when grown in in-vitro propagation.). Due to the possible potential utilization of *Gigaspora margarita* in enhancing food crop production, remediation in polluted soil, and potential fungal evolution when grown in in-vitro propagation culture technique, the researchers decided to explore the influence of different environmental factors that may affect its growth when grown in in-vitro propagation culture technique.

METHODS

This study used an experimental approach, specifically the true experiment design, with the model multigroup design using a quantitative method. The study's setup comprised four groups: one for the control group and three for the experimental group. The researchers extracted *G. margarita* using the wet sieving technique and confirmed its identity through microscopic analysis. The researchers then used potato dextrose agar as their medium and employed a root organ culture where the extracted *G. margarita* was inoculated. The prepared culture was then subjected to varying Temperature, Moisture Level, and Light Intensity conditions, while the control groups were maintained under standard conditions. The three treatment groups were then exposed to different variations;

- Temperature: 30°C, 35°C, 40°C
- Moisture Level: 5%, 10%, 15%
- Light Intensity: 500nm, 600nm, 700nm

The parameters were controlled using heat mats with a temperature thermostat, artificial light with light filters attached to them for light intensity, and the water concentration in the PDA medium for moisture level adjustment. Through data collection which includes quantifying root colonization and hyphal growth rates using ImageJ, a software. The researchers observed the responses of *G. margarita* to different environmental parameters. Statistical analysis, particularly MANOVA, was used to evaluate whether the

growth of *G. margarita* is influenced by the multiple levels of the three environmental factors, either individually or in conjunction with each other. The result of the MANOVA revealed significant variations in growth rates among experimental groups, supporting the hypothesis that environmental factors influence *G. margarita*'s growth.

RESULTS AND DISCUSSION

The researchers propagated *G. margarita* via the in-vitro culture method for seven days after isolating fungus from corn roots. The exposed roots in the three environmental factors were then stained for observation in the microscope. The researchers then measured the growth based on colonization rate and hyphal growth.

Table 1. Growth Rate of *G. margarita*

Parameter	Root Colonization (%)				Hyphal Growth (µm)			
	Mean per group				Mean per group			
	Control	1	2	3	Control	1	2	3
Temperature (Celsius)	0.91	1.09	1.32	1.23	0.01	0.03	0.04	0.01
Moisture Level (%)	2.30	2.31	2.51	2.44	0.01	0.01	0.03	0.01
Light Intensity (nm/lux)	1.22	1.57	1.74	2.06	0.01	0.01	0.03	0.01

Note that the Treatment Groups were as follows: Temperature: 0°C /30°C/ 35°C/ 40°C; Moisture Level: 0%/5%/10%/ 15%; and Light Intensity (500nm, 600nm, 700nm)

As seen in Table 1, the researchers successfully cultivated *G. margarita* in the in vitro propagation technique, in which they achieved the highest root colonization rate at 2.44% and the lowest at 0.91%. Similarly, the recorded hyphal growth in roots ranged from 0.01 µm to 0.04 µm. While optimal levels of root colonization should vary from 3.5% to 96.3% (Carrenho et al., 2007), and the hyphal growth of AMF from 5. 9 µm (when grown after 28 days) to 10.2 µm (when grown for 42 days) if associated with plants, it is important to note that the cultures were propagated at a shorter duration and as Ipsilantis and Sylvia (2007) highlighted, there is currently no standard value for root colonization or hyphal growth within AMF fungi as these values depends on the specific AMF species and its plant partner.



Meanwhile, the researchers tested the null hypotheses from the accumulated data using JAMOV to confirm the observations. The following table presents the p-values computed from MANOVA and the decisions regarding the null hypotheses based on these results:

Table 2. *MANOVA Test Results*

Dependent Variables	Factors	F	F-crit	p	Decision
Root Colonization	Temperature	36.2	6.59	0.002	Rejected
	Moisture	7.22	6.59	0.043	Rejected
	Light Intensity	53.73	6.59	0.001	Rejected
Hyphal Growth	Temperature	12	0.018	0.018	Rejected
	Moisture	9	6.59	0.03	Rejected
	Light Intensity	9	6.59	0.03	Rejected

Table 2 indicates that the p-values associated with various environmental factors are below the significance threshold of 0.05. Rejecting all null hypotheses suggests that all environmental factors significantly influence the growth of *G. margarita* in arbuscular in-vitro propagation. The MANOVA results indicated that moderate temperature (around 35°C) and moisture (approximately 10%) are conducive to the prospering of *G. margarita* in arbuscular in-vitro propagation. These conditions align with findings from studies on other arbuscular mycorrhizal fungal species (Jamiolkowska et al., 2018; Kilpilainen et al., 2020).

CONCLUSION & RECOMMENDATION

The researchers successfully grew *G. margarita* using in-vitro as a method of propagation, in a controlled environment. They observed that maintaining temperature and moisture levels and providing moderate to light intensity led to better colonization rates and hyphal growth in *G. margarita*. These environmental conditions impact the characteristics of the fungus, and higher

levels are linked to faster growth and more desirable traits. Using the in-vitro propagation techniques, the researchers managed to regulate these factors; in-vitro propagation can serve as a preparatory step before introducing *G. margarita* to host plants for improved mutualistic growth.

However, the typical growth rate of AMF species remains uncertain (Bray et al., 2003). This study's results provide insights into conditions that highlight the importance of further research for a comprehensive understanding of *G. margarita* growth patterns. The researchers also recommend investigations to enhance the knowledge about *G. margarita* by refining methods using various media and cultivation techniques and conducting longitudinal studies to observe extended effects. Furthermore, they suggest exploring how *G. margarita* interacts with fungi with its antibacterial effects.

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EFFICACY OF MALUNGGAY (*Moringa oleifera* Lam) SEED GLYCERIN IN MANGO (*Mangifera indica*) PRESERVATION

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Abstract—Mangoes are a widely cultivated fruit in the tropical world. During ripening, mangoes experience various physicochemical changes, including postharvest handling, storage, and climacteric respiration, all of which are susceptible to perishability in certain environmental conditions. Therefore, preserving mangoes is crucial to ensure they are nutritious, tasty, and economical food sources. This necessity has prompted the researchers to look for preservation techniques, eventually leading their investigation into vegetable glycerin utilization. The researchers focused on the potential of glycerin produced by the oil from Moringa Seed, primarily due to its abundance and the fact that it has yet to be well-studied. The researchers employed a multi-group design, observing the shelf life of mangoes in four (4) treatments, each of which had three mango samples. Each group was treated with different Moringa Seed Oil concentrations; thus, the study showed significant results ($p = 0.01 < 0.05$), therefore rejecting the null hypothesis and confirming Treatment B's effectiveness with the composition of (70/25/5%) that was identified as the most effective in extending the mangoes shelf life. Researchers recommend an additional study that tests more mango samples acquired from only one tree or basket to ensure their age and condition are the same.

Keywords: food preservation, malunggay, mango, moringa, vegetable glycerin

INTRODUCTION

Mango is one of the most vital fruits worldwide in producing and consuming countries due to its flavor, taste, and nutritional aspects (Ntsoane et al., 2019). Locally known as "mangga" and scientifically named (*Mangifera indica*), it is the Philippines' national fruit and the third most important fruit crop based on export volume, next to banana and pineapple. Meanwhile, vegetable glycerin is a valuable ingredient in low-sugar food and beverages. Its properties as a sweetener and humectant could help maintain moisture, improve texture, and extend the shelf life of food products (Sharma, 2023). According to Petre (2023), glycerin is a sugar alcohol derived from animal products, plants, or petroleum. On the other hand, glycerin's

safety has been reviewed and confirmed for use in food by health authorities worldwide, including the World Health Organization, the European Union, and countries such as Australia, Canada, and Japan. The U.S. The Food and Drug Administration (FDA) has also declared glycerin safe for consumption (Sollid, 2020).

Therefore, this study aims to preserve mango (*Mangifera indica*) using malunggay (*Moringa oleifera* Lam) seed glycerin as it is known for its susceptibility to rotting, which means that it can spoil quickly and easily. It has been shown to inhibit the growth of bacteria and fungi that cause fruits to spoil, thus extending their shelf life.



The preservation process of Malunggay Seed Glycerin could harness the natural properties of Malunggay to potentially preserve mangoes. Therefore, this study aims to test the efficacy of Malunggay seed glycerin in preserving Mangoes.

METHODS

The researchers employed a true experimental design, specifically a multi-group design, to test the efficacy of malunggay seed glycerin at different concentrations in prolonging the shelf-life of mango. To extract the malunggay seed oil and malunggay seed glycerin, the following materials were used: mortar, pestle, cheesecloth, spatula, pan, saucepan, measuring cups, graduated cylinder, and sterile gloves. Furthermore, a silicon brush was used to apply the malunggay seed glycerin to the mango.

The researchers used the Real-Time Stability Testing method to determine if the Mango is decaying in response to the application of Malunggay Seed Glycerin; the researchers will be observing the Mango (*Mangifera indica*) after it gets coated with Malunggay (*M. oleifera* Lam) seed glycerin at different concentrations every day until it starts to show a sign of rotting. Digital clocks will be used to measure the days and track the progression of preservations. to collect the data of the experiment as accurately as possible.

The researchers used Microsoft Excel and JAMovi to analyze the collected data statistically. They calculated the mean of the days for each mango at different treatments and used the Single Factor ANOVA to analyze the difference between the means of the three independent groups, which is the malunggay seed glycerin at different concentrations specifically; treatment A which is 100% concentration of Pure Malunggay Seed Glycerin, treatment B with 70/25/5%, and treatment C which is 50/45/5% concentration in prolonging the shelf-life of mango. In interpreting the test results, if the p-values were less than $\alpha=0.05$

(level of significance), the researchers would reject the null hypotheses. Given that the null hypotheses are rejected, this indicates that there is a significant difference in the shelf life of mango, where the 70/25/5% concentration is the most effective among the other concentrations. Otherwise, failure to reject the null hypothesis indicates no significant difference in the shelf life of mangoes among the groups.

RESULTS AND DISCUSSION

Different treatments have been used to observe the efficacy of Malunggay Seed Glycerin in prolonging the shelf life of mangoes. After evaluating its potential as a natural preservative, the data gathered from these treatments were statistically analyzed.

Table 1. Comparison of Different Treatments of Mango in Terms of Their Shelf Life

Treatment Groups	Shelf life Per Mango (Days)			Mean
	1	2	3	
A. 100% (Pure Malunggay Seed Oil)	9	9	10	9.33
B. 70/25/5%	11	9	11	10.33
C. 45/50/5	8	8	9	8.33
D. Control Group (No Treatment)	8	8	8	8.00

Note: Formulation Indications: Malunggay Seed Oil/Water/Lye

As seen in Table 1, treatment B had the highest average effectiveness compared to the other concentrations, indicating that mangoes can be preserved with higher concentrations of Malunggay Seed Glycerin. Treatment B, which contains 70/25/5% of Malunggay seed glycerin in mangoes, has shown effectiveness by exceeding 6 more days in the original average shelf life. The researchers observed that other factors, such as temperature, could potentially influence the results of the data collected. Other factors, such as temperature and harvest maturity, may impact the outcome of extending the Mango's shelf life. However, the temperature is the most important environmental factor that influences the deterioration of fruits (Napong & Pavelee, 2021).

The researchers also observed the shelf life of mangoes with different



concentrations of Malunggay Seed Glycerin, comparing how long the shelf life of each mango at different concentrations lasted and how many days it took to exceed the average shelf life. The three trials per treatment group showed different results. Treatment A, a 100% concentration of pure Malunggay seed oil, exceeded the original average shelf life by 5 days. Treatment B, the 70/25/5% concentration of Malunggay seed glycerin, exceeded the original average shelf life by 6 days. Treatment C, the 45/50/5% concentration of Malunggay seed glycerin, exceeded the original average shelf life by 5 days. Lastly, treatment D, which is mango without Malunggay seed glycerin, did not show any change. Malunggay seed glycerin is effective in prolonging mangoes' shelf life based on the data collected. Therefore, it is concluded that the 70/25/5% concentration of Malunggay seed glycerin is the most effective in prolonging the shelf life of mangoes among the other treatment groups.

The researchers further analyzed the differences between the treatments using Single-factor ANOVA to identify the longest shelf-life achieved under the treatments applied, as shown below:

Table 3. Data Collected in Terms of Shelf Life of Mango in Different Concentrations

<i>F</i>	<i>df1</i>	<i>df2</i>	<i>p</i>	Decision
6.67	3	8	0.01	Rejected

Table 3 showed that in terms of effectiveness, using Single-factor ANOVA, the results were obtained using a 0.05 significance level. It is clearly stated in Table 3 that the *F*-computed (6.67) is greater than the *F*-critical (4.06), and the *p*-value (0.01) is less than the alpha (0.05). Thus, the null hypothesis is rejected. Therefore, there is sufficient evidence that the treatments used in determining the effectiveness of the malunggay seed glycerin in prolonging the shelf life of mangoes are significantly different from each other in terms of effectiveness. This confirms that the

Malunggay Seed Glycerin effectively prolongs the shelf life of mangoes, particularly the 70/25/5% concentration.

CONCLUSION & RECOMMENDATION

The effectiveness of malunggay seed glycerin in prolonging the shelf life of mangoes varies depending on the treatment. Various concentrations have shown promising results. Therefore, malunggay seed glycerin can serve as a preservative to enhance the shelf life of mangoes. In particular, Treatment B, containing a specific composition of 70/25/5% of malunggay seed glycerin, has proven to be the most ideal in extending the shelf life of ripe mangoes by over 6 days compared to their original average shelf life. Researchers suggest several recommendations further to optimize the preservative's concentration ratios for greater effectiveness. These include conducting studies on consistent mango samples from a single tree or basket to ensure uniform age and condition. Improving study facilities will yield more outstanding results. Assessing the efficacy of malunggay oil extracts on other perishable fruits, accurately measuring the mangoes' weight daily, and closely monitoring their quality parameters. Precisely measuring the applied malunggay seed glycerin. By adhering to these guidelines, future researchers can advance the understanding and application of malunggay seed glycerin in preserving fruits and contribute to developing sustainable and effective preservation techniques for perishable products.

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SAMBONG (*BLUMEA BALSAMIFERA*) LEAF EXTRACT AS ANTI-TICK SHAMPOO

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Abstract - Ticks (*Ixodida*) are one of the world's biggest public health problems and veterinary problems. They periodically feed with blood, with long intervals between meals. When they bite their hosts, they injure the tissues of animals at their feeding site, causing irritation, inflammation, or hypersensitivity. Sambong (*Blumea balsamifera*) leaf extract has emerged as a potential alternative to anti-tick shampoo. Given the increasing demand for eco-friendly parasitic control alternatives, this study evaluated the efficacy of *B. balsamifera* leaf extract in controlling ticks (*Ixodida*) through laboratory experiments. Further, this study focused on the efficacy of *B. balsamifera* leaf extract as an anti-tick shampoo and determined the most effective concentration for killing ticks. The effectiveness of the anti-tick shampoo made from *B. balsamifera* in different treatments was determined based on the percentage of ticks killed. Thus, the results showed that the treatment containing the highest concentration of *Blumea Balsamifera* leaves extract is the most effective treatment for killing ticks. The results concluded that *B. balsamifera* is an effective alternative component of making anti-tick shampoo.

Keywords: *Blumea balsamifera*, ticks, anti-tick, shampoo

INTRODUCTION

. Linked to spiders since they are arachnids, Ticks (*Ixodida*) are tiny, bloodsucking parasites that come in sizes ranging from that of a pencil eraser to that of a pinhead (Holland & Whelan, 2023). When they bite their hosts, they injure the tissues of animals at their feeding site, causing irritation, inflammation, or hypersensitivity. Massive infections of ticks can cause anemia as a result of blood loss. Moreover, *Ixodida* are one of the world's biggest public health problems and veterinary problems. The economic impact can be broken down into direct and indirect losses and is closely related to the disease's epidemiology. (Hurtado & Giraldo-Ríos, 2019). However, the currently available anti-tick products can contain harmful chemicals that can be bad for animals' skin and coat, leading to skin irritations, redness, itching, and discomfort

for animals. On the other hand, *Sambong*, also known as *Blumea balsamifera*, is a typical medicinal plant across the Philippines (Boy et al., 2018). Additionally, the entire plant, its crude extracts, as well as its isolated constituents exhibit a wide range of biological activities, including those that are antitumor, hepatoprotective, superoxide radical scavenging, antioxidant, antimicrobial, and anti-inflammation, anti-plasmodial, anti-tyrosinase, platelet aggregation, enhance percutaneous penetration, wound healing, anti-obesity, as well as insect and disease resistant (Pang et al., 2014). Thus, the researchers considered making an anti-tick shampoo using *B. balsamifera* leaf extract. This study aims to determine the effectiveness of sambong leaf extract as an anti-tick shampoo in varying concentrations.



METHODS

The researchers used a true experimental research design, which uses a scientific method or approach to establish a cause-effect relationship among a group of variables in a research study. In particular, they used a multi-group design since there were four treatments involved. The researchers prepared 1 kilogram of *B. balsamifera* leaves, 305ml castile liquid soap, a strainer, four 200mL bottles, a bowl, a knife, and a liquid measuring cup. Wearing gloves for safety, they chopped and crushed the *B. balsamifera* leaves and squeezed out the extract with a strainer, and the liquids were measured using the liquid measuring cup based on a desired amount. Then, they poured the liquids into a bowl and stirred using a spoon. The four treatments contained 200mL of sambong leaf extract and castile soap with varying concentrations. The liquids were poured into the 200mL shampoo bottle, and the mixture was set for an hour. Subsequently, the four albino rats were planted with 40 ticks, ten ticks per rat, then let sit for 3 days. The ticks were collected from the researchers' dogs and placed in a container. One albino rat for every treatment, and each rat has a 2-minute bathing time. They regulated the bathing time with stopwatch, scrubbing the rat for 1 minute using the shampoo, gently washing off the shampoo for 30 seconds using tap water, and drying them using a towel. After bathing the rats, the researchers counted the number of tick/s killed. They eliminated the remaining ticks planted. This is repeated four times in every treatment comparing the percentage of number of tick/s killed per each treatment. Moreover, albino rats are used as a test subject. The researchers acquired four albino rats, one albino rat for each treatment with four trials. In selecting the subjects, they make sure that the rats are healthy, 4-10 weeks old, and have no ticks or other parasites. Ticks are collected from dogs with infestations and placed in a plastic container. The ticks that are collected is *Dermacensor variabilis*, specifically at Nymphs stage. There are ten ticks planted on every one

albino rat, a total of 160 ticks are used in experimental setup. The researchers calculated the number of ticks killed by getting the percentage (%) with the formula $\% = (n/N) \times 100$. Where n is the sample size which is the number of tick/s killed and wherein N is the total population which is the total of tick/s planted on each albino rat. And they get the mean of the percentage of the number of ticks killed per group. The average percentage of the number of ticks killed should be greater than or equal to 80% to be interpreted as effective. Meanwhile to compare the test results the researchers will use Analysis of Variance. The researchers tabulated & analyzed the data in JAMOV. The *B. balsamifera* leaves extract will be effective as anti-tick shampoo given that it has higher average percentage of no. of tick/s killed than the castile soap and was confirmed in the ANOVA test results that the null hypothesis should be rejected. If the researchers failed to reject the null hypothesis the *B. balsamifera* leaves extract is not effective as anti-tick shampoo.

RESULTS AND DISCUSSION

The usefulness of *Blumea Balsamifera* leaves extracts as an anti-tick shampoo was tested in this study. Researchers are examining this organic treatment potential. Researchers have demonstrated that extract from *Blumea Balsamifera* leaves effectively kills a particular type of tick specifically *Dermacensor variabilis* by a variety of analyses.

Table 1. Multi-Group Design in Determining the Percentage of Tick/s Killed

Treatments	Number of Tick/s Killed	Percentage (%)
Treatment 1 (Control Group; no Sambong Leaves Extract)	0	0
Treatment 2 (35% Sambong Leaf Extract)	5	50%
Treatment 3 (60% Sambong Leaf Extract)	7	70%
Treatment 4 (100% Sambong Leaf Extract)	9	90%



Based on the data collected, *Blumea Balsamifera* is effective as an anti-tick shampoo, bearing different outcome groups. Treatment 1 with 0% sambong leaves extract and 200% castile soap yielded a result of 0 from the first trial to the fourth trial. In contrast, treatment 2, comprising 65% castile soap and 35% sambong leaf extract, kills 8 ticks in the first trial, 7 ticks in the second, 8 ticks in the second, and 9 ticks in the fourth. On the other hand, treatment 3, which consists of 40% castile soap and 60% sambong leaf extract, killed 10 ticks in the first trial, 7 in the second, and 9 in the third and fourth. Treatment 4 with 100% sambong leaves extract and 0% castile soap killed 9 ticks in the first trial, 10 ticks in the second and third trials, and 8 ticks in the fourth trial. Therefore, treatment 4 has the highest no. of tick/s killed compared to other treatments. According to Pang et al. (2014), the extract of *B. balsamifera* leaves showed a 60.8% insecticidal activity against the adult *Aleurodicus disperses*. Similarly, the extract of *B. balsamifera* leaves effectively eliminates ticks, showcasing its broad-spectrum insecticidal properties.

Meanwhile, Table 2 reveals if *B. Balsamifera* leaf extract is effective as an anti-tick shampoo.

Table 2. Hypothesis Test Results

<i>F</i>	<i>df</i>	<i>p-value</i>	Decision
96.2	12	<0.001	Rejected

As seen in Table 2, the *p-value* is less than 0.05, which means the null hypothesis is rejected, which confirms the researcher's findings. There is a statistically significant difference between the average percentage of tick/s killed in the experimental setup, indicating the effectiveness of *Blumea Balsamifera* leaf extract as an anti-tick shampoo in different varying concentrations.

CONCLUSION & RECOMMENDATION

Blumea balsamifera is effective as an anti-tick shampoo, increasing potency with

higher leaf extract concentration. While the study bore desirable results, further study is necessary to characterize the properties of *Blumea Balsamifera*. Further studies may focus on testing the shampoo further. Researchers could further enhance the results by examining the parasitic characteristics of different *Blumea Balsamifera* parts or incorporating additional components with parasitological properties.

Researchers must work quickly with appropriate authorities, such as environmental advocacy groups, to include this eco-friendly shampoo in local tick control programs, particularly where tick infestations present serious hazards to public health and the economy. It must be used immediately to stop ticks from becoming culturalized.

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EFFECTIVENESS OF TUBA-TUBA (*Jatropha curcas*) SEED EXTRACT AS AN INSECTICIDE AGAINST HOUSEFLY (*Musca domestica*)

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Abstract - There has been a growing interest in finding eco-friendly solutions to control pests and reduce reliance on chemical insecticides. Tuba-Tuba (*Jatropha curcas*) seed extract has emerged as a potential alternative from a natural source. Given the increasing demand for eco-friendly pest control alternatives, this study aims to evaluate the efficacy of Tuba-Tuba seed extract in controlling houseflies (*Musca domestica*) through laboratory experiments. The objectives include assessing the mortality rate of houseflies exposed to different concentrations of Tuba-Tuba seed extract and determining the most effective concentration for controlling housefly populations. Thus, the researchers determined the effectiveness of the insecticide made from Tuba-tuba seeds in the different treatments based on the number of flies killed. The test was performed by putting five flies per container and spraying the solution with various concentrations. The results conclude that the higher the concentration of Tuba-tuba (*J. curcas*) seed extract, the more potent it is in inhibiting house flies.

Keywords: Tuba-tuba, Flies, Insecticide

INTRODUCTION

Flies are known to be vectors for many diseases of both humans and livestock and are considered a sign of unsanitary conditions. Many flies are considered serious pests of livestock, poultry, and equids (Geden et al., 2021). *Musca domestica* (Diptera, Muscidae) is the most common fly in the world. These pests carry pathogens that cause disease (Moran et al., 2022). According to Khamesipour et al. (2018), these pathogens included infant diarrhea, anthrax, cholera, ophthalmia, bacillary dysentery, typhoid fever, and tuberculosis. In addition, houseflies transmitted many of the eggs of helminths such as *Enterobius vermicularis*, *Strongyloides stercoralis*, *Trichuris trichiura*, *Toxocaracanis*, *Taenia*, *Hymenolepis*, *Dipylidium*, *Diphyllobothrium*, and *Trichomonas* species. *Jatropha curcas*, locally known as “Tuba-tuba,” was screened for its toxicity against different

developmental stages of houseflies. The larvicidal, pupicidal, and adulticidal activities were analyzed at various concentrations of hexane and ether extracts (Chauhan et al., 2015). The tuba-tuba (*Jatropha curcas*) has been proven to provide insecticidal properties against flies and their different development. (Chauhan et al., 2015). This study tested the insecticidal properties of *J. curcas* seed extract against houseflies (*M. domestica*). The researchers also seek to understand its efficacy, optimal concentration, and potential applications in pest control. Additionally, it may result in the underlying mechanisms of how the extract interacts with houseflies, providing insights for future development of eco-friendly insecticides.

METHODS

The researchers used a multi-group design in this study. The materials were Pestle and Mortar, which is used to extract the



garlic and *J. curcas* seeds, 60mL spray bottle for the three (3) treatments that will be used to inhibit the flies where the first treatment contains 50% of *J. curcas* seed extract and 50% of Garlic extract. The second treatment contains 75% of *J. curcas* seed extract and 25% garlic extract. The third treatment contains 100% *J. curcas* seed extract. The flies were gathered through improvised traps that the researchers made. The flies were used to test the three treatments to see which concentration was more effective. They were tested through three trials.

The researchers converted the number of flies killed per trial to a percentage by dividing the total number of flies killed by the treatment groups by the total number of houseflies per trial multiplied by 100. They then calculated the mean. They conducted their inferential analysis using Single-Factor ANOVA. They tabulated and analyzed the data using Microsoft Excel.

RESULTS AND DISCUSSION

After collecting data, they converted the number of flies killed per trial and calculated the mean per group. Table 2 summarizes the results:

Table 1. Percentage of Houseflies Killed per Group

Treatment Groups	% of Flies Killed per Trial			Mean (%)
	1	2	3	
A 50% Tuba-tuba, 50% Garlic	20%	0%	20%	13%
B 75% Tuba-tuba, 25% Garlic	20%	40%	20%	27%
C 100% Tuba-tuba 100%	80%	60%	100%	80%

Tuba-tuba and Garlic insecticide can kill houseflies. Of the treatments, Treatment C killed the most houseflies (mean: 80%). The researchers also noticed that the potency of the insecticide increased among treatment groups, which was dependent on the concentration of the extracts. This suggests that the insecticide potency increases with higher concentrations of Tuba-tuba extract. It is also important to remember the limiting

factors that may have played a role in the results, such as the number of test subjects used per trial and how these subjects were tested with the insecticide (Valdez-Ramirez et al., 2023). Tuba-tuba seed extract's insecticidal efficacy may be reduced when mixed with garlic (*Allium sativum*) extract for various reasons. The active chemicals in garlic extract, such as allicin, may interact with the phytotoxins found in tuba-tuba seeds, modifying their insecticidal activity. Garlic extract has been demonstrated to prevent the formation of fungal hyphae, although its insecticidal action has not been thoroughly explored (Ubulom et al., 2021). Furthermore, it may be necessary to identify the ideal concentrations of each extract in order to preserve potency when combined. Perveen and Ahmad (2017) also stated that the combination of entomopathogenic fungi and botanicals has a synergistic effect on the development of houseflies. These findings suggest that the compatibility of garlic and tuba-tuba and their effectiveness against house flies may be influenced by various factors, including their specific properties and the presence of resistance mechanisms in the fly population. Nevertheless, the fact remains that Tuba-tuba can potentially be used as an insecticide, as the results have illustrated.

Meanwhile, the table below summarizes the results of the ANOVA:

Table 2. ANOVA Test Results

<i>F</i>	<i>df</i>	<i>p-value</i>	Decision
16.8	2	0.003	Rejected

The collected data resulted in a *p*-value of 0.003, which is less than the significance level of 0.05. Consequently, the null hypothesis is rejected, confirming the researchers' results.

CONCLUSION & RECOMMENDATION

The experiment's results revealed that the Tuba-tuba Seed Extract has the



potential to be an organic insecticide against houseflies. It becomes more potent at higher concentrations, leading the researchers to conclude that the preferred concentration of the extract should be at maximum (100%). However, there is still room for further improvements. The researchers acknowledged that other insecticidal components could be added to make it more effective.

Further studies may focus on exploring different formulations and combinations of Tuba-tuba seed extract with other natural insecticidal agents to optimize its effectiveness against house flies. They may experiment with various concentrations and application methods to maximize the product's insecticidal properties while ensuring safety for humans, animals, and the environment. They may investigate its efficacy against a broader spectrum of insect pests beyond house flies, including agricultural pests and disease vectors.

Meanwhile, the researchers recommend collaborating with pest management programs and agricultural and environmental agencies to integrate Tuba-tuba seed extract into existing pest management programs and develop guidelines and training materials to educate farmers, pest control professionals, and communities on the proper use and application of Tuba-tuba seed extract for controlling house fly infestations. These actions can help engage the local communities and farmers to promote the adoption of Tuba-tuba seed extract as a sustainable and eco-friendly alternative to synthetic pesticides.

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MAYANA (*Coleus blumei*) LEAVES and DAYAP (*Citrus aurantifolia*) PEEL EXTRACTS as an ALTERNATIVE CONTUSION RUB

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Abstract - In recent years, there has been an increasing interest in exploring alternative remedies using natural resources to address various health conditions. This research focuses on the utilization of Mayana leaves and Dayap peel extracts as an alternative contusion rub. In this study, the researchers used a true-experimental design to investigate the effectiveness of a contusion treatment rub on albino rats containing Mayana and Dayap extracts. Specifically, Solomon Four Group Design was utilized to compare outcomes between groups exposed to different percentages of Mayana and Dayap extracts. The contusion rub made from each treatment had undergone different tests to measure its healing rate. Using the Analysis of Variance (ANOVA), the formulated hypotheses were tested with a probability level of 0.05. After the observations were made, all the data were collected. The researchers concluded that Mayana leaves and Dayap peel extracts can be a viable component in making an alternative contusion rub.

Keywords: albino rats, contusion rub, Mayana leaves, Dayap peel extracts, alternative remedies, healing rate

INTRODUCTION

Contusions, common among athletes and everyday individuals who experience falls or blunt force trauma, are often treated with traditional rubs containing potentially harmful ingredients such as menthol and methyl salicylate (Kumar et al., 2016). While menthol, derived from peppermint, is generally safe, excessive use can lead to adverse effects (Kumar et al., 2016). Mayana and Dayap, traditional herbs in the Philippines, have been scientifically proven to possess analgesic, anti-inflammatory, and antimicrobial properties (Herbanext Laboratories, Inc., 2022). These herbs contain compounds like sterols, triterpenes, and rosmarinic acid, contributing to their therapeutic effects.

Combining extracts of mayana and dayap could provide a natural and safer

alternative to conventional rubs for treating bruises and contusions (Stylecraze, 2023). This study aims to evaluate the healing potential of mayana and dayap extract-based contusion rubs, addressing concerns about the safety of traditional rubs and providing a more sustainable option for managing bruises (Choudhary, 2023).

The researchers conducted a study to explore the potential of Mayana leaves and Dayap peel extracts as an alternative contusion rub. Mayana, scientifically known as *Coleus blumei*, is known for its potential medicinal uses. Studies have shown that Mayana leaves have wound-healing properties. On the other hand, Dayap, also known as *Citrus aurantifolia*, is a citrus fruit whose peels are rich in essential oils and have been used in traditional medicine.

The researchers extracted the active compounds from Mayana leaves and Dayap



peels to develop an alternative natural remedy for contusions. This research fills the gap in the current literature, providing new insights into the potential use of Mayana leaves and Dayap peels in traditional and modern medicine. The study also includes a feature to monitor the effectiveness of the Mayana leaf and Dayap peel extract on contusion healing through a visual measurement method. Further research would be needed to fully understand the effectiveness and safety of this alternative contusion rub.

METHODS

The researchers put 20g of mayana leaves in a basin bowl with 150 mL of olive oil and heated it over low heat for 45 minutes until the oil changed its color. Meanwhile, the researchers mixed the 20g of dayap peels in the basin bowl with 150mL of Olive oil and heated it over low heat for 45 minutes until the oil changed its color. The researchers then filtered the residues from the resulting extracts and combined the oils to be heated again with 20g of beeswax until melted. The final product, 100% Mayana and Dayap (contusion rub) was transferred to a 5g container while still warm. They repeated the procedure for the formulation of the other treatment groups (75% and 50%) but under different amounts of the ingredients used.

The study entailed a multigroup post-test-only design focused on formulating the contusion rub under different treatments. The resulting data was then used to analyze the difference to determine its effectiveness and most potent formulation. Twelve (12) albino rats that met the basic requirements for the experiment were used as target subjects inflicted with contusions of equal sizes in the leg and treated according to the assigned treatments.

The researchers recorded the days until each rat had completely healed per

group. They then compared the data between the groups and tested the null hypotheses using Single Factor ANOVA.

RESULTS AND DISCUSSION

The table below presents the data collected from observing the healing rate of contusions in albino rats with different percentages of Mayana in the contusion rub. The following sections provide a detailed analysis of the outcomes observed during the experiment.

Table 1. The Healing rate of Contusion

Treatment Groups	Healing Rate per Test Subject (Days)			Mean
	1	2	3	
A 100% Extract Concentration	4	5	4	4.33
B 75% Extract Concentration	6	5	5	5.33
C 50% Extract Concentration	7	6	16	6.67
D. Control Group	9	9	9	9

As Table 2 shows, treatment A has the highest average effectiveness compared to the other concentrations; this suggests that increasing the extract concentration increases the potency of the contusion rub. The analysis highlights that Treatment A exhibits the highest average effectiveness compared to other percentages of the Mayana extract. This suggests that the potency of the contusion rub is higher when concentration is increased. Essentially, as the concentration of the mayana extract increases, so does the effectiveness of the treatment in alleviating contusions. This observation implies that higher concentrations of the mayana extract lead to more potent and potentially more beneficial outcomes in treating contusions.

Meanwhile, the table below summarizes the results of the ANOVA:

Table 2. ANOVA Test Results

<i>F</i>	<i>F-crit</i>	<i>p-value</i>	Decision
48.3	4.07	<0.05	Rejected

As indicated in Table 3, the *p*-value (0.00) is less than the alpha level (0.05), indicating that the null hypothesis in this variable is rejected. This confirms that the



Mayana is effective in healing contusions, particularly the rub with 100% percentage.

CONCLUSION & RECOMMENDATION

The researchers formulated a contusion rub from the extracts of Mayana and Dayap. The Mayana Leaves and Dayap Peel Extracts are effective in treating a contusion. The rub with 100% Mayana worked the best, meaning the higher the percentage of Mayana extract, the more effective the rub is. The contusion rub can be applied to albino rats as it is proven to heal contusions, but there is still a need to study the further adverse effects of the contusion rub on the skin.

The researchers recommend focusing on the significance of the amount of Mayana leaves concentrated in the contusion rub, growing Mayana in a controlled environment, and studying the alternative use of pure extract instead of decoction. They also recommend using Dayap peels in the data extract instead of its fruit. Another scope for skin and allergy testing is to test the contusion rub on other test subjects, such as albino rats, to ensure safety. Professionals should also supervise contusion rub testing. To innovate other aspects, a study on the antioxidant and antifungal properties of Oregano and Dayap should be considered. Since the current study was conducted within a specific timeframe, future researchers should involve long-term observation and evaluation to obtain a more realistic and accurate understanding of this combination. This approach would allow researchers to evaluate the efficacy and longevity of the treatment method over an extended period.

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HEALING RATE OF GUMAMELA (*HIBISCUS ROSA-SINENSIS*) AND VIRGIN COCONUT OIL (*COCOS NUCIFERA L.*): OINTMENT FOR OPEN WOUND TREATMENT

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Abstract - This study explored the potential of gumamela (*Hibiscus rosa-sinensis*) as an ointment for promoting wound healing in albino mice. The research involved applying a mixture of gumamela extract and virgin coconut oil to open wounds of albino mice and comparing their healing progress. The albino mice were divided into three groups: those treated with different amounts of gumamela, 100 grams, 50 grams, and no gumamela. The wound-healing process was monitored by measuring the rate of wound contraction, assessing the appearance of the wound, and evaluating the overall health of the mice. A statistically significant difference in wound healing does not exist between the groups. The three groups displayed comparable rates of wound contraction and overall wound appearance. Additionally, no adverse effects were observed in either group during the study period. The different amounts of gumamela in ointment do not exhibit a significant advantage in promoting wound healing in albino mice. Further studies may be necessary to investigate alternative concentrations, application methods, or additional natural components that could enhance the wound-healing properties of this combination.

Keywords: Gumamela, Virgin Coconut Oil, Wound Healing, Albino Mice

INTRODUCTION

Bacteria are known for being everywhere, giving the world infectious surroundings, resulting in humans creating things that will help the world find an easy way to overcome them. Throughout history, humans have relied on plants as vital resources, extracting herbs and compounds for medicinal purposes (Bassal et al., 2020). Botanical products and herbal supplements are recognized for treating diseases and promoting good health, leading to the widespread use of herbs in various forms of complementary medicine (Ewumi, 2022; Herbal Medicine, 2021). Skin conditions, including burns and wounds, present

persistent challenges in treatment, prompting the need for innovative wound management solutions (Stan et al., 2021). This study investigates the efficacy of an oil-based ointment in promoting fast wound healing. By utilizing coconut oil and gumamela extract through the maceration extraction method, the researchers aim to develop an ointment with enhanced wound-healing properties (Lima et al., 2015). This ointment is expected to accelerate wound healing and possess antibacterial properties, which are crucial for preventing wound infections and complications (Boateng, 2016). Furthermore, using locally available components like coconut oil and gumamela ensures that the



product is cost-effective and easily accessible, particularly in the Philippines. The research investigates combining alternative therapies and plant extracts for antibacterial, anti-inflammatory, and wound healing properties, advancing knowledge alternative healthcare and sustainable medicine. Additionally, this research may investigate the healing rate and efficacy of an ointment based on gumamela in the treatment of open wounds.

METHODS

This study employed a true-experimental design, specifically utilizing a Post Test Only Multi-Group Design. This approach involves comparing outcomes among distinct groups that undergo varied treatments or interventions to establish cause-and-effect relationships between the variables under investigation. Regarding data collection, the researchers utilize tools like razors and sterilized blades to make precise incisions on mice, along with measuring tape to quantify wound sizes and weighing scales to monitor the health of the mice. The study focused on six mice carefully chosen for their uniformity in size, weight, and overall health, minimizing potential sources of variability. Data analysis entailed tracking healing rates and employing ANOVA tests to evaluate the effectiveness of different ointment formulations. The primary objective is to reject the null hypothesis, indicating that there is a discernible effect resulting from the intervention. This comprehensive approach ensures rigorous examination of the research question and contributes to the body of knowledge in wound healing.

RESULTS AND DISCUSSION

The researchers conducted their experiment by applying three distinct ointments, each containing varying amounts of Gumamela Extract, to three groups of mice. Subsequently, they meticulously observed and recorded the healing rates exhibited by each group of mice over a

specified period. This methodological approach allowed the researchers to comprehensively analyze the potential effects of gumamela extract dosage on wound healing in their experimental subjects, providing valuable insights into its therapeutic efficacy. The following sections provide a detailed analysis of the outcomes observed during the experiment.

Table 1. Comparison of The Healing Rate of Test Subjects Under Different Treatments of Gumamela and Virgin Coconut Oil Ointment

Treatment Groups	Healing Rate per Test Subject		Mean
	1	2	
A. 100% Extract Concentration	0.100mm ²	0.050mm ²	0.0750mm ²
B. 50% Extract Concentration	0.083mm ²	0.050mm ²	0.0665mm ²
C. 0% Extract Concentration	0.071mm ²	0.050mm ²	0.0605mm ²

Based on the experiment, the result showed that regardless of whether the ointment was applied or not, the mice showed similar levels of improvement. This indicates that the ointment did not have any noticeable effect on the outcome of the experiment. While statistical analysis may not reveal a significant difference in healing rates between the treatments, factors such as the lack of groups, environmental influences, and the absence of other natural healing ingredients may contribute to varying rates of healing, even if they do not manifest as statistically significant differences across groups. Thus, considering adding another group for treatment beyond statistical averages can provide valuable insights into the effectiveness of interventions.

Table 3. Single Factor ANOVA Test Results

F	F-crit	df	p-value	Decision
0.15	9.55	2	0.86	Not Rejected

Table 3 confirms the comparison of the groups. With a p-value of 0.86, the researchers failed to reject the null hypothesis. Hence, there is no statistically significant difference on the wound healing



rate of the mice on the treatments, indicating the ineffectiveness of *Gumamela* flower extract and Virgin Coconut Oil as wound ointment. The lack of statistical disparity among the three treatments could be attributed to factors such as an insufficient number of treatment groups or errors in data collection. A more comprehensive examination is necessary to elucidate the reasons behind the absence of clear statistical outcomes.

CONCLUSION & RECOMMENDATION

This study delved into utilizing *Gumamela* and Virgin Coconut Oil as a potential natural remedy for treating open wounds, aiming to investigate its maximum effectiveness by comparing the product to different treatments. However, despite extensive analysis, the researchers could not establish conclusive evidence supporting its effectiveness in treating wounds since all mice tested have similar wound healing rates. Factors such as the size of the wound, the absence of critical natural healing components, intricacies involved in wound recovery, and constraints within the study design might have influenced the outcomes. Nonetheless, this research contributes to the ongoing exploration of natural remedies for open wound management, emphasizing the necessity for further investigation in this domain.

Future studies could explore the effects of different *gumamela* extract concentrations, virgin coconut oil uses, and care methods on albino mice to optimize the ointment's efficacy. Long-term observation and evaluation are necessary to understand the treatment's effectiveness and longevity over an extended period.

Comparative studies with other treatments can provide valuable insights into the efficacy and safety of *gumamela* and virgin coconut oil as an ointment for albino

rats. Finally, expanding the study to include larger participant groups, diverse concentrations, or modified settings could lead to further understanding of its potential effects.

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Enhancing the Mycorrhizal Symbiosis of Chili Plant (*Capsicum Frutescens*) Using Rinsed Rice Water

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Abstract - There has been a growing interest in finding eco-friendly solutions to enhance or improve the growth and the characteristics of planting, even other crops. Mycorrhizal Symbiosis of Chili Plant (*Capsicum frutescens*) using Rinsed Rice Water has emerged as a potential alternative for watering plants, as it is obtained from a natural source. Given the increasing demand for eco-friendly enhancement, this study evaluated the efficacy of Rinsed Rice Water in enhancing Chili plants through post-test only control group design. Chili Plants watered with rinsed rice water had better plant growth and development ((Average Number of Leaves: 23.20; Height: 38.80 cm; Root-Shoot Ratio: 1.64 cm; and Dry Weight: 4.10 g).) than those that were not (ilan mean), hinting that the intervention has implications in symbiosis between the plants and the mycorrhizal fungi. Further studies are necessary and should focus on such implications.

Keywords: AMF, Chili, Mycorrhizae, Agriculture



INTRODUCTION

Chili plant (*Capsicum frutescens*) cultivation was a significant agricultural practice in the Philippines, particularly in regions like Bicol, where the indigenous *Siling Labuyo* variety thrives. Renowned for its fiery heat, *Siling Labuyo* was emblematic of Filipino cuisine and held cultural significance nationwide. This resilience made them accessible for cultivation, even in domestic settings, and reflected the widespread practice of home gardening among Filipinos. Successful chili pepper farming entails several critical steps, from seedling production to transplanting, as Braganza (2022) outlined. These steps were vital for ensuring robust growth and optimizing yields, thus influencing the economic viability of chili farming ventures.

Moreover, it underscores the economic potential of chili farming, citing instances where market prices for *Siling Labuyo* reached notable levels, particularly in regions like Batangas. However, success in chili farming requires a comprehensive understanding and management of various factors, including optimal planted seasons, irrigation practices, and pest-controlled strategies. Furthermore, the dynamic nature of chili markets, characterized by fluctuating prices and evolving consumer preferences, underscores the importance of strategic planning and informed decision-making among farmers. The characteristic of the said fruit is rounded with a pointed tip, uncooked versions are green, and matured versions are crimson, smoothed, medium-sized, elliptical left and thin, and 30-120cm long branched distinguishing features. This fruit contains vitamins, minerals, and antioxidants (Braganza, 2022). In light of these considerations, this study aimed to investigate the key determinants of successful chili pepper farming in the Philippines. By examining factors such as agronomic practice, market dynamics, and socio-economic factors, this research seeks to provide insights that could enhance the efficiency and profitability of chili farming operations. Additionally, the study aimed to contribute to the broader discourse on sustainable agriculture and rural development in the Philippines, emphasizing the significance of chili pepper farming within this context.

The symbiotic association between planted and arbuscular mycorrhizal (AM) fungi was widely considered beneficial for both. However, a significant amount of ecological studies highlight how context-dependent this interaction was. Different abiotic variables, such as nutrition availability, could affect interactions along the mutualism-parasitism continuum and have distinct effects on a plant's ability to grow and maintain its fitness—the benefits of AM fungus for planting extended beyond improved phosphorus feeding and enhanced plant development. For example, although fungi were considered parasitic regarding food availability, the organism in question demonstrates its importance by improved tolerance to both biotic and abiotic stresses. Little was known about the benefits that AM fungus got from plants, except that biotrophs were necessary (Chen, 2018).

According to Jaylea (2022), carbohydrates and healthy nutrients are presented in the rinsed rice water. In addition to giving the planted carbohydrates that it could store in its cell membrane for use as fuel, the starches in rice water also encouraged the growth of beneficial bacteria and other microorganisms that were already presented in the soil, such as lactobacilli and mycorrhizae, by giving them a source of nourishment. Furthermore, because washed rice water contained many carbohydrates, which acted as a natural fertilizer booster for all plants, it remarkably influenced plant growth and development. In small amounts, the washed water also contained several nutrients and minerals like potassium, phosphorus, and nitrogen that could be used to nourish Lactobacilli and other soil microorganisms (Shields, 2023). Rinsed rice water may also have a good effect on the symbiotic fungus by fostering an environment that was more conducive to their growth, as these nutrients helped the soil microflora, which in turn supported mycorrhizal growth. Hence, this study tested whether such a hypothesis was true, studying its effect on mycorrhizal symbiosis by evaluating the growth of the chili plants with the application of rinsed rice water.



METHODS

The researchers used a posttest-only group design in this study where there were 2 groups of Chili comprising 10 seedlings planted with AMF: A) The Experimental Group, which the researchers watered with rinsed rice water, and B) The Control Group, which they watered with regular water. After 5 weeks, they uprooted the chili plants from the soil to measure the following parameters: Number of Leaves, Height, Root-Shoot Ratio, and Dry Weight. They calculated the mean of the groups on the said parameters and compared the results using an independent samples t-test.

RESULTS AND DISCUSSION

The table below summarizes the results of the experiment:

Table 1. Results of the Experiment

Mean per Plant Characteristic Parameter				
Group	Average Number of leaves	Height (cm)	Root-Shoot Ratio (cm)	Dry Weight (g)
A (Experimental Group)	23.20	38.80	1.64	4.10
B (Control Group)	10.20	30.6	0.98	1.94
t-Value	6.78	2.66	2.69	2.31
p-Value	0.0003	0.045	0.027	0.049
Decision	Rejected	Rejected	Rejected	Rejected

Based on the collected data, the plants from the experimental group yielded better results in all parameters, indicating that rinsed rice water effectively enhanced the chili plant's mycorrhizal symbiosis. Group A (Experimental) had a bigger size and larger number of leaves, longer lengths, bigger roots, and more heavy dry weight. Group B (Control) had a smaller size and lesser number of leaves, shorter lengths, lesser and smaller roots, and a lighter dry weight. Therefore, the experimental group that had rinsed rice water was more effective than the control group with regular water.

Furthermore, all the null hypotheses for the parameters were rejected, which confirms the researcher's findings. Rinsed rice water indeed enhances the mycorrhizal symbiosis between the chili plant and the AMF, leading to better plant growth.

CONCLUSION & RECOMMENDATION

The researchers were able to study the effect of rinsed rice water on the mycorrhizal symbiosis of the Chili Plant by observing the plant's growth and development. Rinsed rice water enhances the relationship, leading to improved growth and development in Chili Plants. Immediate collaboration between researchers and relevant authorities, such as agricultural or environmental agencies, urgent actions are needed to integrate this eco-friendly solution into other harmful fertilizers, especially for significant health and economic risks.

Further studies may still be conducted to explore how mycorrhizal fungi can interact with plants that do not typically have symbiotic relationships. This exploration could enhance the



mycorrhizal efficiency while minimizing harm to the environment. Future researchers may also add another week of observation that contains chili qualities.

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COMPARATIVE ANALYSIS OF D-LIMONENE EFFICIENCY OF CITRUS SUBSPECIES

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Abstract – Citrus fruits are known for their valuable components, especially d-Limonene, widely used in various industries. This study aimed to compare the efficiency of d-Limonene extraction among lemon (*Citrus limon*), sweet orange (*Citrus sinensis*), and pomelo (*Citrus maxima*) through improvised steam distillation. The objective would be to determine the extraction time and yield of d-Limonene from each citrus subspecies. Data showed significant differences in extraction time ($p = 0.037$) and d-Limonene yield ($p = 0.031$) between citrus subspecies. Lemon had the most balanced extraction time and yield of d-Limonene, followed by pomelo and sweet orange. This study adds to our understanding of optimizing d-Limonene methods for obtaining from citrus fruits, with implications for industrial uses and agricultural practices. By identifying lemon as a greater source of d-Limonene, companies can modify their extraction procedures to enhance productivity and efficiency, making better use of citrus resources. Furthermore, these findings may influence agricultural methods, assisting farmers in selecting citrus cultivars for increased d-Limonene yield. Overall, this study closes the information gap in citrus-based d-Limonene extraction, having implications for both industry and agriculture.

Keywords: Citruses, d-Limonene, Steam Distillation, Extraction

INTRODUCTION

d-Limonene, a widely occurring terpene found abundantly in citrus oils, particularly in oranges, lemons, and pomelo, holds significant importance due to its diverse range of applications and potential health benefits. It is extensively utilized in various industries, including food flavoring, fragrance production, and healthcare, where it is believed to offer therapeutic properties against ailments like bronchitis, cancer, and diabetes (Nikfar et al., 2014). Moreover, d-Limonene plays a crucial role in the industrial sector, serving as a key ingredient in manufacturing hand sanitizers, plant pesticides, and chemical solvents. Furthermore, the diversity of citrus fruits, including lemon, orange, and pomelo, provides a rich source of d-Limonene. This

comprehensive technique enables us to compare d-Limonene content and extraction yields across different citrus fruits, providing useful insights into the best sources for d-Limonene extraction.

A comparative analysis of D-limonene efficiency among citrus subspecies, including Lemon (*Citrus limon*), Orange (*Citrus sinensis*), and Pomelo (*Citrus maxima*), lies in the need to understand the variations in d-Limonene yield and extraction Time efficiency across different citrus varieties. Each citrus subspecies may possess unique biochemical compositions and peel characteristics that could influence the extraction process and ultimately impact d-Limonene production efficiency. By systematically comparing these subspecies, researchers can identify which citrus



varieties offer the highest yield of d-Limonene, optimize extraction techniques tailored to specific subspecies, and inform industries seeking to maximize the utilization of citrus fruit byproducts. Additionally, this comparative analysis contributes to the broader understanding of citrus diversity and highlights potential opportunities for enhancing the sustainability and profitability of citrus cultivation and processing industries.

METHODS

This study employed a comparative approach to test the efficiency of d-Limonene extraction among various citrus subspecies through a multi-group design, which allowed for a systematic comparison of d-Limonene yield across different citrus varieties, and each of these citrus subspecies represents a separate experimental group. Moreover, the independent variable is the type of citrus subspecies, including lemon (*Citrus limon*), sweet orange (*Citrus sinensis*), and pomelo (*Citrus maxima*). The dependent variable currently under testing is the d-Limonene yield obtained through steam distillation. Researchers prepared 300g of different citrus fruit peels: lemon, orange, and pomelos. They started the distillation process by grating the samples, mixing them with water, and heating the apparatus at low temperatures. The extraction time was recorded using a stopwatch, and the yield was measured using a graduated cylinder. The procedure was repeated three times per citrus subspecies being tested. After the experiment, all equipment and glassware were cleaned and properly stored, and waste materials were disposed of according to laboratory safety guidelines. The researchers ensured the safety of their participants. The researchers used ANOVA and JAMOVI to analyze data on extraction time and d-Limonene yields from various citrus subspecies. They calculated the mean per trial and used Single Factor ANOVA to compare results. If the null hypothesis is rejected, the extraction time and yield that

produce the quickest and highest amount of d-Limonene are the most suitable for d-Limonene production. If the null hypothesis is rejected, all samples are viable for production, with high yield and minimum extraction time.

RESULTS AND DISCUSSION

The researchers calculated the mean of each group to determine which group produced the highest d-Limonene Yield. The table below shows the result.

Table 1. d-Limonene Yield and Extraction Time per Group in each Citrus Subspecies.

Subspecies	Extraction Time			Mean	Extraction Yield (mL)			Mean
	(Minutes) per Trial				per Trial			
	1	2	3		1	2	3	
Lemon	37	47	55	46.33	21	26	28	25.00
Orange	35	56	60	50.33	23	28	31	27.33
Pomelo	30	43	45	39.33	21	25	25	23.67

As seen in Table 1, The extraction process of d-Limonene from citrus fruits is a complex process that involves various factors such as the quality and ripeness of the fruits, variations in extraction techniques, and genetic differences among citrus subspecies. The mean extraction time, which represents the average time taken to complete the extraction process across all trials within a particular subspecies group, varies among different citrus varieties. For example, lemons take 37, 47, and 55 minutes to extract d-Limonene, while oranges and pomelo require 50.33 and 39.33 minutes, respectively. The mean extraction yields, representing the average volume of d-Limonene obtained from the extraction process, also vary among different citrus varieties. These variations in yield suggest inherent differences in the D-Limonene content or extraction efficiency among the various citrus varieties studied. Further investigation into optimization strategies for



enhanced yield and consistency in industrial applications is necessary to address these complex interplays (Siddiqui et al., 2022). Satira et al. (2021) highlighted that d-Limonene, a renewable resource from citrus peel oil, is used in various industries like food, beverage, pharmaceutical, and biopesticides as an antibacterial agent. Its non-toxic nature makes it a viable biological alternative in bioengineering and a starting substrate for forming molecules, including p-cymene, an intermediate in industrial catalytic processes. Demand for limonene exceeds supply.

Table 2. ANOVA Test Results

Parameter	F	df	p	Decision
Extraction Time	9.48	2	0.04	Rejected
Extraction Yield	10.31	2	0.031	Rejected

Table 2 demonstrates that the p-values for different extraction times and yields are less than 0.05. All of the null hypotheses were rejected; this signifies a statistically significant difference in the extraction time of the D-limonene Citrus subspecies and a statistically significant difference in the D-limonene Yield of the Citrus subspecies. This demonstrates that lemons are more efficient and effective in terms of time and yield than pomelo and oranges. Factors such as the quality and ripeness of the fruits, variations in extraction techniques, and genetic differences among citrus subspecies could contribute to this observed variability (Rizzo et al., 2023)

CONCLUSION & RECOMMENDATION

This study analyzed D-limonene production from three citrus subspecies: lemon, orange, and pomelo. Oranges had longer extraction times but higher yields, while pomelos were faster but slightly lower. Lemon was the most balanced option, with a moderate extraction time and yield. Oranges were practical due to their adaptability, as

their peels can extract d-Limonene, reducing waste and providing an ecological option. All tested citrus varieties proved viable for d-Limonene production. However, the study suggests that future research should include phytochemical analysis to identify d-Limonene in plant extracts, using technologies like GC-MS to detect and quantify concentration (Sahnoun et al., 2023). The study also suggests investigating the effect of different extraction times on juice output, particularly for lower-yielding fruits like lemons, to optimize extraction and maintain quality. Researchers should also investigate different extraction methods for each citrus type (lemon, orange, pomelo) to optimize d-Limonene yield, as oranges produce more under similar conditions. This could significantly improve overall output.

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Utilization of Nile Tilapia (*Oreochromis Niloticus*) Scale as Roof Sealant

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Abstract - In response to the need for sustainable solutions for roof quality, an increasing need to explore more sustainable and innovative solutions has been increasing. Because Nile Tilapia scales are rich in hydroxyapatite, collagen, and calcium carbonate, the researchers looked into its potential use in making a durable and environmentally friendly sealant to address roof leakage. This research focuses on the utilization of Nile Tilapia Scale, a waste material from the fish industry, into a durable and environmentally friendly roof sealant. The researchers formulated roof sealants out of different concentrations of Nile Tilapia Scales (50%, 75%, 100%, and 0%) combined with resin at corresponding amount, and tested the products in multi-group experimentation regarding their sealing capacity, adhesiveness, chemical reactivity, and thermal resistance. While pure Nile Tilapia Scales as roof sealant is ineffective, it can be mixed with resin at a 50% concentration, enhancing the sealing capacity, adhesiveness, thermal resistance, and chemical reactivity of the resulting roof sealant. Further studies may also be necessary, particularly in improving the sealant's thermal resistance to withstand warmer climate conditions of tropical countries like the Philippines.

Keywords: roof sealant, tilapia scales, resin



INTRODUCTION

Roof leakage presents a persistent challenge to housing quality, infrastructure maintenance, and building longevity, impacting both occupants and the built environment (Shan & Zhou, 2016; Pelchen & Benningfield, 2022). Traditional roof sealants are crucial barriers against environmental elements, safeguarding homes from water infiltration and structural damage (Walczak & Scalisi, 2022). However, innovative solutions are imperative to address this ubiquitous problem effectively.

Recent exploration into alternative materials has revealed the potential of fish scales as a viable option for roof sealants (Akter et al., 2017). Tilapia scales, in particular, offer a compelling opportunity to mitigate waste in the fishing industry while providing a sustainable solution to roofing challenges (Akter et al., 2017). Composed of a blend of organic and inorganic components, including collagen fibers and minerals like hydroxyapatite and calcium carbonate, tilapia scales possess characteristics conducive to durable and long-lasting roof sealants (Laurens & Fletcher, 2022).

The utilization of tilapia scales in roof sealants addresses the pressing need for effective leak prevention and contributes to environmental sustainability by repurposing a byproduct of the fishing industry (Akter et al., 2017). This study aims to investigate the feasibility and efficacy of tilapia scales as a roofing material, exploring their sealing capacity, adhesiveness, thermal resistance, and chemical reactivity. By leveraging the unique properties of tilapia scales, this research seeks to provide a practical and eco-friendly solution to roof leakage, enhancing the comfort and resilience of homes against severe weather conditions such as rain and heat.

METHODS

The researchers used a multi-group design in this study. They boiled the Nile Tilapia Scales and mixed the resulting paste with resins at varying amounts depending on the treatments. The first treatment contained 50% of the tilapia scales, Treatment 2 with 75%, and Treatment 3 contained 100%. Lastly, as the control group, Treatment 4 did not have tilapia scales but only resin in the roof sealant formulation. The sealants were then applied to a roof sample and let dry for 24 hours. The researchers then tested the sealants according to four parameters: Sealing Capacity, Adhesiveness, Thermal Resistance, and Chemical Reactivity.

In evaluating the effectiveness of the roof sealants, the researchers quantitatively observed the performances on the parameters using a validated structured observation sheet comprised of Likert scales. They calculated the means per trial and compared the groups in all the parameters under the Multi-Analysis of Variance (MANOVA).

RESULTS AND DISCUSSION

The table below summarizes the results:

Table 1. Experimental Results

Group	Mean per Parameters			
	Sealing Capacity	Adhesiveness	Thermal Resistance	Chemical reactivity
A (50%)	4.50	4.50	3.33	4.80
B (75%)	3.90	3.40	2.60	4.40
C (100%)	1.00	1.00	1.00	1.00



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D (Control, 0%)	2.60	2.60	2.90	1.60
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Performance of the Treatment Groups per Parameter

Note: 1.00-1.80 = Very Weak; 1.81-2.60 = Weak; 2.61-3.40 = Average; 3.41-4.20 = Strong; and 4.21-5.00 = Very Strong

As seen in Table 2, Treatment A always gets the highest mean scores in all parameters (Sealing Capacity: 4.50; Adhesiveness: 4.50; Thermal Resistance: 3.33; and Chemical Reactivity: 4.8). The researchers also noticed that the Treatment Groups increase in scores in the parameters the lesser the concentration of the fish scales in the roof sealant formulation was used. Although Fish Scales alone are inefficient as roof sealant (Sealing Capacity: 1.00; Adhesiveness: 1.00; Thermal Resistance: 1.00; and Chemical Reactivity: 1.00), it is also important to note that the rest of the treatment groups performed better than the control group which used pure resin (Sealing Capacity: 2.60; Adhesiveness: 2.60; Thermal Resistance: 2.90; and Chemical Reactivity: 1.6). This indicates that the Nile Tilapia Scales is good as an additive component in making roof sealant which increases the formulation's effectiveness, achieving maximum performance with 50:50 Ratio of the Scales and the resin. According to Aziz et al. (2021) fish scales, often a by-product of the fish market, can be repurposed to create durable items with unique mechanical qualities. Composed of collagen fibers and calcium-deficient hydroxyapatite, similar to bone and teeth, they exhibit distinct hardness, fracture strength, and elastic modulus. Researchers have successfully developed a composite material using fish scales and a natural organic binder, indicating the potential for innovative applications. In addition, the transformative properties of fish scales, composed primarily of collagen and calcium phosphate, which become adhesive under heat exposure, with collagen identified as the critical component (Gonzalez et al., 2021). This research indicates that Nile Tilapia Scales are effective as an additive component in enhancing the effectiveness of the formulation of nile tilapia scale as roof sealant, achieving maximum performance with a 50:50 ratio of scales and resin. Overall, the research findings demonstrate the potential use of fish scales as part of roof sealant, providing protection against harsh weather conditions and extending the roof's lifespan. The findings underlined the importance of conducting thorough research and development on natural resources for industrial applications.

Meanwhile, of all the parameters, the groups yield the least in terms of thermal resistance, suggesting the need for possible additive components to enhance the roof sealant on the said aspect, especially because the Philippines is a tropical country, and sealants must be resilient in warm temperatures. According to Tukel et al. (2019), the results show that the best thermal insulation performance is obtained with an air gap width of 9 mm for all situations investigated, while the poorest performance is obtained with a gap width of 6 mm or 15 mm depending on the number of panes. This indicates that the thicker the roof sealant, the stronger the thermal resistance. Thus, the ability of the enclosing envelope to delay heat transfer through the building structure over time is determined primarily by the thickness and type of construction material utilized, which determines the envelope's massing. It is an additional crucial factor in determining the building's thermal performance and, consequently, the amount of energy needed to maintain thermal comfort in the inhabited area (Al-Homoud, 2015).

Finally, the researchers confirm the results through the MANOVA test results below:

Table 3 MANOVA Test Results

Parameter	F	df	p-Value	Decision
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Sealing Capacity	54.8	3	<0.01	Reject
Adhesiveness	69.1	3	<0.01	Reject
Thermal Resistance	38.9	3	<0.01	Reject
Chemical Reactivity	128.7	3	<0.01	Reject

Table 3 clearly states that the p-values (<0.001) are lesser than the alpha (0.05) on all parameters, rejecting the null hypotheses. There is sufficient evidence that the treatments used in determining the effectiveness of the Nile Tilapia scales as roof sealant are significantly different in terms of effectiveness, confirming the researchers' interpretation.

CONCLUSION & RECOMMENDATION

The researchers successfully created a roof sealant out of Nile Tilapia Scales and have tested its full potential. Although ineffective on its own, it can be added with resin at a 50% concentration, enhancing the sealing capacity, adhesiveness, thermal resistance, and chemical reactivity of the resulting roof sealant. Further studies may also be necessary, particularly in enhancing the sealant's thermal resistance to withstand warmer climate conditions of tropical countries like the Philippines.

Recommendations

While consumers can readily use the current formulation of the roof sealant, the researchers encouraged further study of the efficacy of Tilapia Scales as a roof sealant to explore and enhance its use in formulating a roof sealant, further enhancing its commercial viability.

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